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A novel boundary-integral algorithm for viscous interaction of deformable drops

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A new three-dimensional boundary-integral algorithm for deformable drops moving in a viscous medium at low Reynolds numbers is developed, which overcomes some familiar difficulties with boundary-integral calculations. The algorithm is used to simulate different modes of interaction between drops or bubbles, primarily for buoyancy-driven motion. The present iterative method for mean curvature calculation is found to be more robust and accurate than contour integration schemes. A novel iterative strategy based on combining biconjugate gradient and simple iterations overcomes the poor convergence of “successive substitutions” for drops in very close approach with extreme viscosity ratio. A substantially new variational method of global mesh stabilization solves the problem of mesh degradation with advantageous, soft stability constraints. A curvatureless boundary-integral formulation is also derived and shown to provide, in principle, a more accurate description of the drop breakup than the conventional formulation. The efficiency of these techniques is demonstrated by numerical examples for two drops in gravity-induced motion with high surface resolutions. The present code successfully simulates mutual approach of slightly deformable drops to extremely small separations, as well as their rotation when in “apparent contact,” thus bridging the gap between finite deformation calculations and a recent asymptotic theory for small capillary numbers. Also provided is a 3D simulation of the experimental phenomenon of enhanced bubble coalescence, discovered by Manga and Stone [*J. Fluid Mech.* **256**, 647 (1993); **300**, 231 (1995)]. For drops of viscosity comparable to that of the surrounding fluid, it is shown in contrast that breakup is a typical result of hydrodynamic interaction in gravity-induced motion for large and even moderate capillary numbers. The code is readily applicable to any type of an ambient flow and may be adapted to more than two drops. © 1997 American Institute of Physics. [S1070-6631(97)00506-0]

I. INTRODUCTION

Low Reynolds number motion of deformable drops is a fundamental problem of relevance to emulsion sedimentation and rheology, and to drop coalescence and breakup. For zero capillary numbers, $Ca = 0$, when the drops remain spherical, numerous studies have been performed in recent years to investigate hydrodynamic interactions,^{1–9} collision efficiencies,^{4,9,10–12} and rheological and sedimentational properties of semi-dilute¹³ and moderately concentrated¹⁴ emulsions. For $0 < Ca \ll 1$, deformation can be considered as a regular^{15–18} or singular^{19,20} perturbation; most recently, we have applied the singular perturbation approach to a 3D problem of drop coalescence in gravity-induced motion.²¹ When $Ca = O(1)$, the deformation is no longer a small perturbation, and a numerical approach based on boundary-integral methods becomes appropriate. This approach, pioneered for deformable drops by Rallison and Acrivos,²² is described in detail by Pozrikidis.²³ The applications to axisymmetrical problems have included the motion of a drop toward a solid wall or an interface^{24–28}, drop breakup in an extensional flow,^{29,30} development of instabilities for a single

drop in gravity-induced motion,^{31,32} gravity-induced motion of two drops,³³ deformation and breakup of contaminated drops,^{34,35} and the thermocapillary migration of two drops.³⁶ Three-dimensional solutions have included the steady-state deformation and the onset of a breakup for a single drop in a shear flow,³⁷ rheological calculations for a dilute emulsion in a shear flow,³⁸ transient motion of ordered emulsions,³⁹ and the interaction of two or more drops.⁴⁰ Most recently, considerable progress has been made by Loewenberg and Hinch,⁴¹ who developed a boundary-integral dynamical simulation code for calculating rheological properties of random moderately concentrated emulsions in a shear flow and performed calculations with up to 12 equisized drops in a periodic cell. Novel features of their technique include a “near-singularity” subtraction, to improve the convergence of the boundary-integral discretization for drops in close approach, and adding artificial “springs” between the mesh points to stabilize the grid and facilitate long-time calculations. They considered⁴¹ a range of parameters (drop volume fraction ϕ and drop-to-medium viscosity ratio λ) where the effect of interactions on the bulk stress is still less than the single drops contribution; additional studies are necessary to determine how this code handles quite strong hydrodynamic interactions, which we are particularly interested in.

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Also relevant to the above mentioned studies are very interesting experiments by Manga and Stone^{33,40} on buoyancy-driven motion of two bubbles in a highly viscous corn syrup; most strikingly, they showed that strong deformation may lead to the smaller bubble being swept around the larger one and then sucked into it, so that the capture efficiency (the maximum horizontal offset far upstream leading to capture) even exceeds Smoluchowski's value for non-interacting drops. The existence of the sucking-in phenomenon was also qualitatively confirmed by their two-dimensional simulations for cylindrical drops. The accompanying 3D boundary-integral calculations for interacting drops in buoyancy-driven motion,⁴⁰ however, have been limited to $\lambda = 1$ (instead of $\lambda \ll 1$, as appropriate for bubbles) and to only the initial stage of approach, and thus could not predict whether the sucking-in phenomenon or drop breakup takes place.

Somewhat apart from boundary-integral simulations is the front-tracking method of Unverdi and Tryggvason⁴² for single and interacting bubbles and drops. A particular strength of this method is the ability to consider a finite-Reynolds number case. For this advantage, the method pays by the presence of the outer computational boundary and a finite interface thickness; the effects of these artifacts, especially the latter, on the solution should be assessed. The limit of creeping flow, which is considered in the present paper, is difficult,⁴² and the results of Unverdi and Tryggvason are essentially for moderate and moderately high Re where the physics may be different. It appears therefore impossible to make comparisons between this method and boundary-integral calculations.

An accurate 3D boundary-integral simulation for deformable drops with arbitrary viscosity is a significant numerical task with intrinsic difficulties. The difficulties with the coordinate singularities, inherent to global surface parametrization, are overcome by using schemes with uniform surface triangulation. However, reliable mean curvature calculation is known to be a problem, which particularly affects the simulations of breakup and merging, like those discussed in Sec. IV. Another familiar difficulty is the mesh degradation in dynamical simulations. The idea of adding artificial "springs" between the nodes, with some "grid tension," which was found to successfully control the mesh for drops freely suspended in a shear flow,⁴¹ turns out to be too inefficient for drops in gravity-induced motion; because of much stronger mesh degradation for the latter case, this stabilization method allows only extremely small numerically stable time steps for fine resolutions, thus making only crude calculations possible in practice. Another problem is the very poor convergence of traditional "successive substitutions" for iterative solution of boundary-integral equations in the case of drops in very close approach and with extremely small or extremely large viscosity ratio λ , even after Wielandt's deflation^{43,23} has been made. This poor convergence has a mathematical origin in the spectrum of the boundary-integral operator being nearly continuous for small surface separations, and hampers not only bubble-bubble coalescence simulations at very small λ , as is typical for Manga and Stone's experiments,^{33,40} but also boundary-

integral simulations for solid particles ($\lambda = \infty$) in close approach.

The present work seeks to overcome the above difficulties and offers a number of new ideas in boundary-integral simulation. The specific goal of the paper is to develop a highly accurate and efficient code for two deformable drops in gravity-induced motion, for further applications in the systematic study of bubble or drop coalescence and breakup. This problem is chosen as being more difficult numerically and less studied than drop motion in shear flows. However, our code may be applied to drops exposed to an ambient flow, as an example in Sec. III E shows, and can be extended to the case of $N > 2$ drops, as long as we are satisfied with a method which has $O(N^2)$ scaling. The direct generalization for the case of N drops in a periodic cell, with $O(N^2)$ scaling, may be also made, if the nearest image contribution is subtracted from the periodic Green function and the remainder is tabulated prior to dynamical simulations—an efficient idea used by several authors in different contexts (see, e.g., Refs. 44,41).

In Secs. II and III A, the boundary-integral formulation and general details of the numerical technique are summarized. Section III B presents a new, iterative method of calculating the curvature and normal vectors on a triangulated surface. This algorithm is a further development of Rallison's³⁷ idea and is based on locally fitting the best paraboloid by least squares. We show through model examples (Sec. III B) and additional boundary-integral calculations (Sec. III E) that the new algorithm of fitting a paraboloid to calculate curvature is more robust and preferred, at least for the present applications, than is contour integration. In Sec. III C, a new iterative strategy based on combining biconjugate gradient and simple iterations is described, which can provide, in practice, as much as an order-of-magnitude saving in the number of iterations for drops in very close approach with extreme viscosity ratio $\lambda \ll 1$ or $\lambda \gg 1$. The novel method described in Sec. III D for global mesh stabilization is based on adding a tangential velocity field found iteratively at each time step to minimize, on the average, the rate of change of the distances between neighboring mesh nodes. Our code using these methods is tested in Sec. III E on a problem of two equisized drops suspended in a shear flow. Most importantly, for gravity-induced motion (Sec. IV), our mesh stabilization algorithm has much softer stability constraints than the method based on grid tension and allows 1–2 orders-of-magnitude larger time steps, making feasible calculations with high surface resolutions. In Sec. III F, we show how the troublesome mean curvature term can be removed altogether from the 3D boundary-integral formulation. For $\lambda = 1$, this new, curvatureless form is easily regularized and leads to a more accurate numerical solution than the conventional boundary-integral formulation, if the drops are not too close. Further work is needed, however, to make this new boundary-integral formulation useful for $\lambda \neq 1$ and/or drops in close approach. Section IV presents several physically interesting and numerically challenging examples of the 3D gravity-induced motion of two drops. First, we show that the present code allows us to follow the mutual approach of slightly deformable drops to extremely small

separations, as well as the subsequent motion in “apparent contact,” thus extending boundary-integral calculations to the range of $Ca \ll 1$ where the singular perturbation theory of drop deformation and coalescence²¹ is already applicable. Section IV also presents an adequate 3D boundary-integral simulation of sucking-in and enhanced coalescence phenomena discovered by Manga and Stone,^{33,40} with an extremely small viscosity ratio $\lambda = 10^{-3}$ typical for bubbles in a highly viscous medium. At the same time, we show that, if the bubbles are replaced with drops having $\lambda = O(1)$, then hydrodynamic interactions in gravity-induced motion may instead lead to drop breakup even for moderate capillary numbers. In Sec. V, we outline some particular advantages of the new numerical technique and discuss other possible applications involving problems of emulsion and suspension mechanics.

II. BOUNDARY-INTEGRAL FORMULATION

Consider two deformable drops of density ρ' and viscosity μ' settling in an unbounded quiescent medium of density ρ_e and viscosity μ_e at negligibly small Reynolds numbers. The indirect boundary-integral approach for deformable interfaces, which has become standard after Rallison and Acrivos,²² gives the system of boundary-integral equations for the interfacial velocity $\mathbf{u}(\mathbf{y})$:

$$u_\ell(\mathbf{y}) = \sum_{\alpha=1}^2 2\kappa \int_{S_\alpha} \mathbf{u}(\mathbf{x}) \cdot \boldsymbol{\tau}_{\mathbf{n}(\mathbf{x})}^{(\ell)}(\mathbf{x}; \mathbf{y}) dS_\alpha + F_\ell(\mathbf{y}), \quad \ell = 1, 2, 3, \quad (1)$$

where

$$F_\ell(\mathbf{y}) = \sum_{\alpha=1}^2 \frac{2}{\mu_e(\lambda + 1)} \int_{S_\alpha} f(\mathbf{x}) \mathbf{n}(\mathbf{x}) \cdot \mathbf{G}^{(\ell)}(\mathbf{x} - \mathbf{y}) dS_\alpha, \quad (2)$$

$$\kappa = \frac{\lambda - 1}{\lambda + 1}, \quad \lambda = \frac{\mu'}{\mu_e}, \quad (3)$$

S_α are the drop surfaces,

$$\mathbf{G}^{(\ell)}(\mathbf{x} - \mathbf{y}) = G_\ell \mathbf{e}_m = -\frac{1}{8\pi} \left[\frac{\mathbf{e}_\ell}{r} + \frac{r_\ell \mathbf{r}}{r^3} \right] \quad (\mathbf{r} = \mathbf{x} - \mathbf{y}) \quad (4)$$

is the fundamental solution, i.e. the Stokeslet velocity generated at $\mathbf{x}$ by the unit force applied at $\mathbf{y}$ along the ℓ th Cartesian axis (assuming unit viscosity),

$$\boldsymbol{\tau}_{\mathbf{n}(\mathbf{x})}^{(\ell)}(\mathbf{x}; \mathbf{y}) = \tau_{ij}^{(\ell)}(\mathbf{x} - \mathbf{y}) n_j(\mathbf{x}) \mathbf{e}_i \quad (5)$$

and

$$\tau_{ij}^{(\ell)}(\mathbf{x} - \mathbf{y}) = \frac{3}{4\pi} \frac{r_i r_j r_\ell}{r^5} \quad (6)$$

are the corresponding stress vector on the interface and stress tensor for the fundamental solution, $\mathbf{n}(\mathbf{x})$ is the outward unit normal at $\mathbf{x}$, and $\mathbf{e}_i$ is the unit vector of the i th Cartesian axis. Finally,

$$f(\mathbf{x}) = 2\sigma k(\mathbf{x}) + (\rho_e - \rho')gz, \quad (7)$$

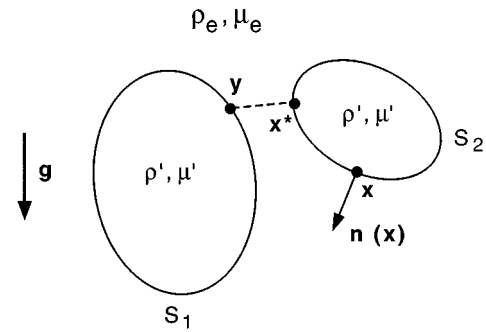


FIG. 1. Schematic of two deformable drops settling under gravity (not to scale).

$f(\mathbf{x})\mathbf{n}(\mathbf{x})$ is the stress jump across the interface (see, e.g., Pozrikidis³¹) with the modified pressure $p - \rho g z$, $k(\mathbf{x}) = \frac{1}{2}(k_1 + k_2)$ is the mean surface curvature at $\mathbf{x}$, $\sigma = \text{const}$ is the surface tension, g is the gravity acceleration, and z is the Cartesian coordinate in the direction of gravity (Fig. 1). For further references, the pressure associated with (4) and (6) is^{23,43}

$$q^{(\ell)}(\mathbf{x} - \mathbf{y}) = -\frac{r_\ell}{4\pi r^3}. \quad (8)$$

It is also useful to note that, for drops freely suspended in an ambient flow $\mathbf{u}_\infty(\mathbf{x})$, Eq. (1) holds with $2\mathbf{u}_\infty(\mathbf{y})/(\lambda + 1)$ added to $\mathbf{F}(\mathbf{y})$ and $\rho_e = \rho'$ in (7).

It is well-known²³ that arbitrary rigid-body motions $\mathbf{u}(\mathbf{x})|_{S_\alpha} = \mathbf{V}_\alpha + \boldsymbol{\Omega}_\alpha \times \mathbf{x}$ are the eigensolutions of (1) with $\kappa = 1$, and uniform expansions $\mathbf{u}(\mathbf{x})|_{S_\alpha} = \mathbf{n}(\mathbf{x})$ are the eigensolutions of the adjoint system with $\kappa = -1$. For these reasons, the numerical solutions of (1) with $\lambda \ll 1$ or $\lambda \gg 1$ suffer from ill-conditioning and are difficult to obtain by iterations because of the slow convergence (see also additional comments in Sec. III C). Wielandt's deflation^{23,43} considerably improves the situation by eliminating $\kappa = \pm 1$ from the spectrum of the integral operator (1). The essence of this procedure is to replace (1) with an equivalent system

$$\mathbf{w}_\ell(\mathbf{y})|_{S_\alpha} = \kappa \left[2 \int_{S_1 + S_2} \mathbf{w}(\mathbf{x}) \cdot \boldsymbol{\tau}_{\mathbf{n}(\mathbf{x})}^{(\ell)}(\mathbf{x}; \mathbf{y}) dS_\alpha - \mathbf{w}'(\mathbf{y}) + \frac{n_\ell(\mathbf{y})}{S_\alpha} \int_{S_\alpha} \mathbf{w} \cdot \mathbf{n} dS \right] + F_\ell(\mathbf{y}), \quad \alpha = 1, 2 \quad (9)$$

for $\mathbf{w} = \mathbf{u} - \kappa \mathbf{u}'$, where the prime denotes the projection of a vector field on the space of rigid-body motions (see below) and then find

$$\mathbf{u} = \mathbf{w} + \frac{\kappa}{1 - \kappa} \mathbf{w}'. \quad (10)$$

Surprisingly, although Wielandt's deflation has been known for some time, only Loewenberg and Hinch,⁴¹ to the best of our knowledge, report having used this powerful tool in practical boundary-integral calculations for deformable drops.

The rigid-body projection $\mathbf{w}'(\mathbf{x})$ can be found without the Gram-Schmidt orthogonalization, simply as

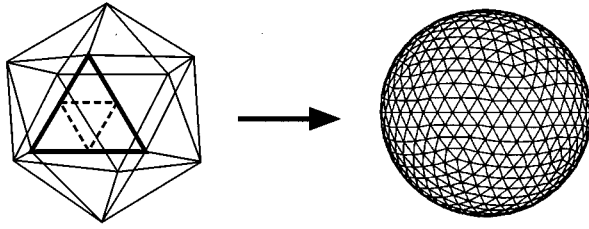


FIG. 2. An example of the surface triangulation into 1280 elements starting from a regular icosahedron (method I).

$$\mathbf{w}'(\mathbf{x})|_{S_\alpha} = A_i \mathbf{e}_i + B_i \mathbf{e}_i \times \hat{\mathbf{x}} \quad (\hat{\mathbf{x}} = \mathbf{x} - \mathbf{x}_\alpha), \quad (11)$$

where

$$\mathbf{x}_\alpha = \frac{1}{S_\alpha} \int_{S_\alpha} \mathbf{x} dS \quad (12)$$

is the surface centroid,

$$A_i = \frac{1}{S_\alpha} \int_{S_\alpha} w_i(\mathbf{x}) dS \quad (i=1,2,3) \quad (13)$$

and the coefficients $B_i (i=1,2,3)$ are calculated from the solution of a 3×3 system

$$D_{ij} B_j = \int_{S_\alpha} \mathbf{w}(\mathbf{x}) \cdot [\mathbf{e}_i \times \hat{\mathbf{x}}] dS \quad (14)$$

with a positive-definite matrix

$$D_{ij} = \int_{S_\alpha} [\delta_{ij} \hat{\mathbf{x}}^2 - \hat{x}_i \hat{x}_j] dS. \quad (15)$$

III. NUMERICAL METHOD

A. General details

There are two general approaches to the surface discretization: (i) global parametrization (see, e.g., Ref. 53) and (ii) uniform triangulation. The second approach is used in the present work, to avoid complications due to the coordinate singularities. The drops start from spherical shapes, and several slightly different schemes were used to construct a sphere triangulation. In the first method,⁴³ each face of a regular icosahedron inscribed into the sphere is divided into four triangles by the edge midpoints, the new vertices are projected radially on the sphere and the process is repeated as many times as necessary (Fig. 2). The second method differs only in that we start from a regular dodecahedron, project each pentagon face center radially on the circumscribed sphere, connect the projection with the face vertices, and then proceed like in the first method. These tessellations give, somewhat restrictively, the total number N_Δ of triangles per drop of the form $20 \cdot 4^k$ or $60 \cdot 4^k$. In several runs, a more flexible approach⁴¹ was used to divide a drop surface into $N_\Delta = 20n^2$ triangles by simply fragmenting each icosahedron face into n^2 equal triangles and projecting the new vertices radially onto the circumscribed sphere. The last method, however, gives less uniform initial triangulations;

for example, with $N_\Delta = 2420$, the maximum-to-minimum mesh edge ratio is 1.41, while for the first two schemes this ratio is always within 1.19. Regardless of the method, all $N_\Delta/2 + 2$ vertices on a drop have coordination number 6, except for only 12 vertices with coordination number 5.

Equations (9) are approximated as a linear system for unknowns $\mathbf{w}(\mathbf{y}_i)$ at the triangle vertices (the only collocation points). Of particular importance is the preceding singularity and “near-singularity” subtraction by the identities

$$\begin{aligned} 2 \int_{S_\alpha} \mathbf{w}(\mathbf{x}) \cdot \boldsymbol{\tau}_{\mathbf{n}(\mathbf{x})}^{(\prime)}(\mathbf{x}; \mathbf{y}) dS_{\mathbf{x}} \\ = 2 \int_{S_\alpha} [\mathbf{w}(\mathbf{x}) - \mathbf{w}(\mathbf{x}^*)] \cdot \boldsymbol{\tau}_{\mathbf{n}(\mathbf{x})}^{(\prime)}(\mathbf{x}; \mathbf{y}) dS_{\mathbf{x}} \\ + \begin{cases} 0 & \text{for } \mathbf{y}_i \text{ outside } S_\alpha \\ \mathbf{w}(\mathbf{y}_i) & \text{for } \mathbf{y}_i \in S_\alpha \end{cases} \end{aligned} \quad (16)$$

and

$$\begin{aligned} \int_{S_\alpha} f(\mathbf{x}) \mathbf{n}(\mathbf{x}) \cdot \mathbf{G}^{(\prime)}(\mathbf{x} - \mathbf{y}) dS_{\mathbf{x}} \\ = \int_{S_\alpha} [f(\mathbf{x}) - f(\mathbf{x}^*)] \mathbf{n}(\mathbf{x}) \cdot \mathbf{G}^{(\prime)}(\mathbf{x} - \mathbf{y}) dS_{\mathbf{x}}, \end{aligned} \quad (17)$$

where $\mathbf{x}^*$ is the collocation node on S_α that is nearest to $\mathbf{y}_i$. For $\mathbf{y}_i \in S_\alpha$ (when $\mathbf{x}^* = \mathbf{y}_i$), the Eqs. (16)–(17) represent the usual singularity removal technique (Pozrikidis,²³ p. 178); for $\mathbf{y}_i$ not on S_α , they were introduced into boundary-integral calculations by Loewenberg and Hinch.⁴¹ In our simulations (Sec. IV), with a relatively large number of elements, we have found these transformations to be successful in providing the solutions for drops in very close approach. Note, however, that, unlike the true singularity removal making the integrands $O(1)$, the subtraction of “near-singularity” does not completely eliminate the unbounded behavior of the double-layer integrand in (16) for drops in close approach and large N_Δ ; so, the important question remains whether this technique can be further improved.

The right-hand side integrals (16)–(17), as well as (12)–(15), are approximated by a simple surface trapezoidal rule that requires the integrands only at the triangle vertices. This rule can be conveniently written, for any function $\varphi(\mathbf{x})$, in the following form, by reassigning the triangle contributions to vertices:

$$\int_{S_\alpha} \varphi(\mathbf{x}) dS \approx \sum_i \varphi(\mathbf{x}_i) \Delta S_i, \quad (18)$$

$$\Delta S_i = \frac{1}{3} \sum \Delta S, \quad (19)$$

where the summation in (19) is over all flat triangle areas ΔS with the vertex $\mathbf{x}_i$. Using (18)–(19) greatly expedites the evaluation of the integrals (16)–(17), the most time-consuming part. Rallison³⁷ was, to our knowledge, the first to practice the economy (18)–(19).

The system (9) is solved iteratively for $\mathbf{w}(\mathbf{y}_i)$, with a linear extrapolation from the two previous time steps as an initial approximation. In updating the drop shapes, a suitable

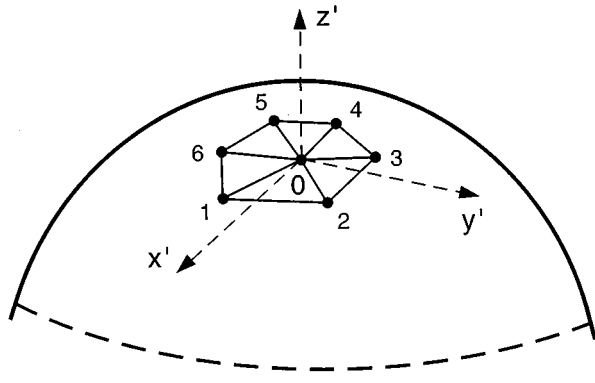


FIG. 3. Schematic for the calculation of the mean curvature and normal vector at point O of a triangulated surface by the best paraboloid fitting, for the typical coordination number of six.

tangential field $\mathbf{\Pi}(\mathbf{x})$ is added to the interfacial velocity $\mathbf{u}(\mathbf{x})$ to stabilize the meshes, and the dynamic equations $d\mathbf{x}_i/dt = \mathbf{u}(\mathbf{x}_i) + \mathbf{\Pi}(\mathbf{x}_i)$ for the collocation points are solved by the second-order Runge–Kutta method. Some aspects of our algorithm (normal vector and curvature calculation, the choice of the iterative method, mesh stabilization) warrant the additional discussion below.

B. Mean curvature and normal vector calculations

Our algorithm of calculating the mean curvature $k(O)$ and outward normal $\mathbf{n}(O)$ at a vertex O of a triangulated surface (Fig. 3) is based on the following simple idea. If the z' -axis of the local Cartesian coordinates (x', y', z') with the origin O was directed along the normal vector $\mathbf{n}(O)$, then z' as a quadratic function of x' and y' would be a good local representation of any smooth surface. However, $\mathbf{n}(O)$ is not known *a priori*, and so the method is iterative. First, a paraboloid is found with a chosen direction of the z' -axis, which passes through O and fits best its neighbors by the least-squares method. The unit normal to this paraboloid at O is used as the new direction of the z' -axis, and the procedure is repeated until the convergence is reached. Eventually, $\mathbf{n}(O)$ and $k(O)$ for the surface S_α are taken as the corresponding quantities for the fitted paraboloid.

The formal description of the algorithm is as follows. Let $\mathbf{n}_0$ be an initial approximation to the outward unit normal at O .

- (1) Choose local Cartesian coordinates (x', y', z') with $O = (0, 0, 0)$ and the z' -axis along $\mathbf{n}_0$. Find (x', y', z') -coordinates of the adjacent nodes.
- (2) Find the coefficients A, B, C, D , and E which minimize

$$F = \sum_{i=1}^{N_O} \frac{(Ax'_i + By'_i + Cx_i'^2 + Dx'_iy'_i + Ey_i'^2 - z_i')^2}{(x_i'^2 + y_i'^2 + z_i'^2)}. \quad (20)$$

Here the summation is over all N_O adjacent nodes, and the weights $(x_i'^2 + y_i'^2 + z_i'^2)^{-1}$ are used to make closer points more important.

- (3) Set $(\mathbf{n}_0)_{\text{new}} = (-A, -B, 1)/(1 + A^2 + B^2)^{1/2}$ in the coordinate system (x', y', z') , as the unit normal to the paraboloid at O .
- (4) If $\|(\mathbf{n}_0)_{\text{new}} - \mathbf{n}_0\| > \epsilon$ go to 1.
- (5) Mean curvature $k(O) = -(C + E)$.

In step 1, the choice of (x', y', z') is determined to within arbitrary rotations about the z' -axis, but this freedom does not affect the resulting $\mathbf{n}(O)$ and $k(O)$. In dynamic simulations, $\mathbf{n}_0$ can be always taken from the preceding step, and in practice the convergence is extremely fast, with only a few iterations needed, even for $\epsilon \sim 10^{-6} - 10^{-8}$. A typical coordination number $N_O = 6$ exceeds the number of unknowns A, B, C, D and E , making the use of the least square method in step 2 quite natural; in rare cases $N_O = 5$, the best paraboloid in step 2 passes through all the adjacent nodes (Obviously, for $N_O \leq 4$ additional close nodes could be used to make the method work, but this is not necessary for the present simulations.) The solution of the symmetric system $\partial F / \partial A = \dots = \partial F / \partial E = 0$ for the unknowns A, B, C, D and E by Gaussian elimination is quite economical, and so the calculation of normals and mean curvatures by this algorithm takes a negligible portion of the total computation time, even for the smallest number of triangles used.

A more common method of calculating the mean curvature in boundary-integral simulations is based on the well-known relation

$$k(O)\mathbf{n}(O) \approx \frac{1}{\Delta S} \int_{\Delta S} k \mathbf{n} dS = -\frac{1}{2\Delta S} \oint_C \mathbf{t} ds, \quad (21)$$

where C is a small contour on the surface enclosing point O , ΔS is the surface area inside C , and $\mathbf{t}$ is the unit outward normal to C lying in the tangential plane. However, we did not find contour integration sufficiently accurate for the present purposes, and so performed several tests to compare the paraboloid fitting algorithm and the contour integration method of obtaining local curvatures. In these tests, the contour integration scheme was implemented as described in Ref. 41, with the provision for some triangles to be obtuse, when the contour construction slightly complicates. In this scheme, the contour C is chosen as the path that passes through the bisectors of the triangle edges surrounding O , and the $\mathbf{t}$ -vectors are taken tangent to the triangle faces (Fig. 4).

The contour integration was found to be much better than paraboloid fitting in the simplest test of a spherical surface with unit radius, its maximum error in k being 3.5×10^{-5} for $N_\Delta = 1280$, compared to 6.6×10^{-3} for our algorithm; however, the curvature variation, of primary importance in view of the regularization (17), is within 3×10^{-4} for our method in this case, which is quite acceptable. In a more crucial set of tests, with prolate ($a=1, b=c=0.4$) and oblate ($a=b=1, c=0.4$) spheroids and a 3D ellipsoid ($a=1, b=0.7, c=0.5$) described by $x^2/a^2 + y^2/b^2 + z^2/c^2 = 1$, we used from 320 to 20 480 triangles, the mesh refinement procedure I (as described in Sec. III A) and the simplest mapping $(x, y, z) \rightarrow (ax, by, cz)$ to obtain ellipsoid triangulations from unit sphere triangulations. Typical results of the comparison between the two methods

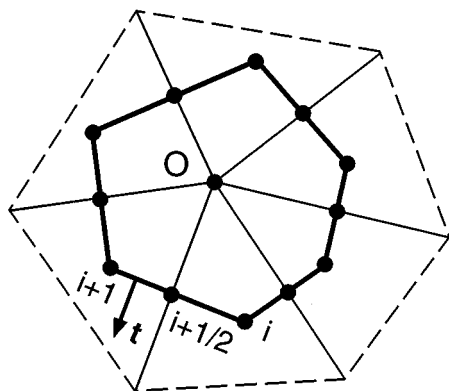


FIG. 4. Schematic for the mean curvature calculation at point O of a triangulated surface (acute triangle case) by the contour integration (21). As recommended (Ref. 41), the contour C (bold) passes through the bisectors of the triangle edges, and $\mathbf{t}$ is a unit vector tangent to the triangle faces and perpendicular to C ; for the line segments $(i, i+1/2)$ and $(i+1/2, i+1)$, this implies that $\mathbf{t}$ is the unit vector of the corresponding mesh edge.

are given in Fig. 5. On the average [Fig. 5(a)], the contour integration method is almost competitive for a crude mesh $N_{\Delta}=320$, but becomes at least several times less accurate than the present algorithm for fine resolutions. More significantly, the maximum error [Fig. 5(b)] of our method is much smaller even for crude meshes and systematically tends to zero at $N_{\Delta} \rightarrow \infty$; in contrast, the contour integration method does not show pointwise convergence at $N_{\Delta} \rightarrow \infty$. This pointwise non-convergence of the contour integration scheme is significant: in the prolate spheroid test of Fig. 5, for example, there are about 25 mesh points with $|k - k_{\text{exact}}| > 0.15$, both for $N_{\Delta}=1280$ and 5120. Similar results were obtained for the oblate spheroid and the 3D. The largest difference between the two methods in pointwise behavior was observed in the oblate spheroid test, with the maximum absolute error in the curvature being 0.051 for the present method vs. 0.57 for contour integration at the typical value of $N_{\Delta}=1280$. In each of these triangulations, some of the triangles are obtuse, but we observed the same problem of pointwise non-convergence with the contour integration method even for a nearly spherical shape ($a=1$, $b=0.85$, $c=0.7$) when all of the mesh triangles are acute. This non-convergence does not correlate with the coordination number and occurs for all three methods of the unit sphere triangulation discussed in Sec. III A.

In the contour integration scheme, the unit normals $\mathbf{n}(O)$ can be found simply by normalizing the integrals (21), with proper sign, and we have found this procedure to provide an accuracy comparable to that of our method in the spheroid and ellipsoid tests in the wide range $N_{\Delta} \leq 5120$. Instead of non-convergence, a slow decrease $\sim N_{\Delta}^{-1/2}$ of the maximum error in the normals for the contour integration method was observed, compared to $O(N_{\Delta}^{-1})$ for our algorithm; this difference, though, was significant only for $N_{\Delta} > 3000$. However, in nearly flat regions or transitional regions (where the mean curvature changes sign), contour integration may not be reliable for calculating normals, be-

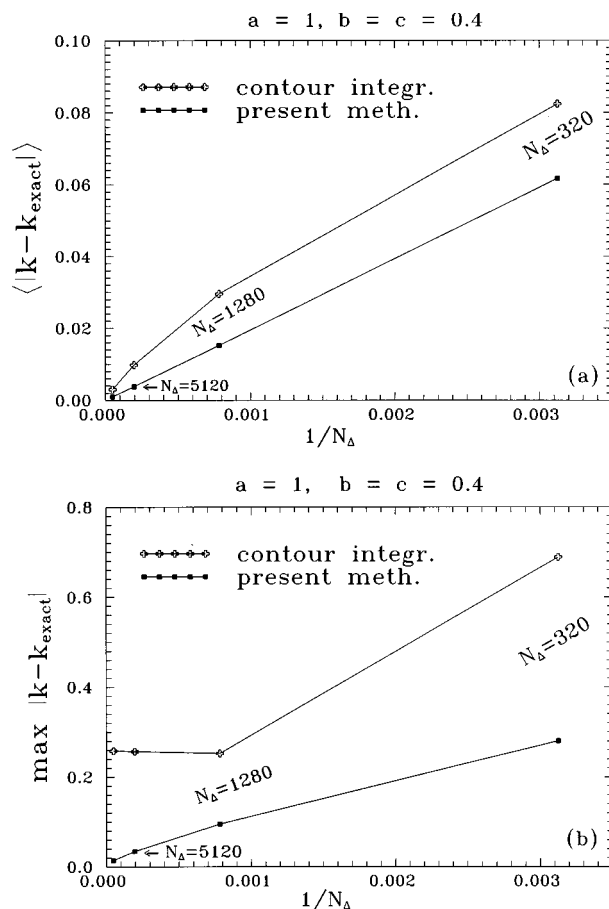


FIG. 5. The average (a) and maximum (b) absolute errors in the mean curvature calculation for a prolate spheroid ($a=1$, $b=c=0.4$) and a different number N_{Δ} of triangle elements, from 320 to 20 480 (mesh refinement method I). Crosses are for the contour integration, and squares are for the present method.

cause of the small values of the integrals in (21), and should be necessarily complemented by an alternative method. For example, we performed tests with the axisymmetrical surface obtained by rotating the curve

$$r(\theta) = \left[1 + 15 \sin^2 \left(\frac{\theta}{2} \right) \right]^{0.4r} \left[1 + \cos^2 \left(\frac{\theta}{2} \right) \right]^{-1} \quad (22)$$

about the $\theta=0, \pi$ axis, which has a dimple near $\theta=0$ and is nearly spherical away from $\theta=0$. Surface triangulations were simple radial projections of triangulations for a unit sphere centered at the origin. With this shape, contour integration fails to provide convergent normals in the dimple region for $N_{\Delta} < 5120$, whereas the present method based on fitted paraboloids converges smoothly with an $O(N_{\Delta}^{-1})$ error on the entire surface. At $N_{\Delta}=5120$, the maximum error in the unit normal is 0.01 radian for paraboloid fitting and 0.09 radian for contour integration. Note that dimples and nearly flat spots are typical in real boundary-integral calculations for bubbles and drops in close approach (Sec. IV). Presumably, accurate normals are more important in the problem for drops in buoyancy-driven motion than for drops freely suspended in a shear flow, at least, for $\lambda = O(1)$, because in the specific case $\lambda = 1$, the boundary-integral formulation of the second problem requires only the product $k(\mathbf{x})\mathbf{n}(\mathbf{x})$. (See

Sec. II.) In the mean curvature test for the dimpled shape (22), we have found contour integration to be more accurate on the main, nearly spherical part of the surface, but less accurate in the dimple region, especially for high resolution; pointwise non-convergence of the contour integration method is much less pronounced or even absent in this case.

Overall, we have found that the contour integration method gives an attractively small error for nearly spherical surfaces, but may lead to unreliable results in other situations. In contrast, the present algorithm of paraboloid fitting is always robust for smooth surfaces and is usually more accurate. The pointwise non-convergence for the contour integration method was found to be strongly dependent on the mapping. If the simplest mapping $(x, y, z) \rightarrow (ax, by, cz)$ of a unit sphere on an ellipsoid is replaced by radial projection from the common center:

$$(x, y, z) \rightarrow \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} \right)^{-1/2} (x, y, z), \quad (23)$$

the non-convergence problem disappears or becomes much less pronounced; the curvature errors, however, in this case are much larger than those for our method in the above tests. The pointwise non-convergence of the curvature in the contour integration method can be removed altogether, if the contour C is chosen differently as the dashed-line perimeter in Fig. 4, with the $\mathbf{t}$ -vectors averaged over two adjacent triangles (separated by a dashed line); this makes the method, however, less local and *always* much less accurate than the present algorithm.

Among known previous algorithms for the curvature and normal vector calculations, Rallison's³⁷ is the closest to our method. In his method, however, the surface is locally given as $f(x, y, z) = 0$, not using the preferred orientation of the z -axis, and up to quadratic terms in x, y , and z are retained to satisfy $f(x, y, z) = 0$ for the central point O and five *ad hoc* neighbors. This leads to a more complicated, non-linear system of equations for nine unknowns solved by Newton–Raphson iterations. Our algorithm requires the solution of a symmetric linear system for five unknowns at each iteration, being an order-of-magnitude faster. We have not attempted a comparison between our curvature algorithm and the more complicated Rallison's method, but the accuracy of the two approaches is expected to be generally similar, since they are based on close principles. However, the use of least squares in our algorithm is designed to serve a better robustness for arbitrary triangulations. Another recent algorithm³⁸ uses quadratic boundary elements but is not aimed directly at calculating the curvature at the triangle vertices, and so we have not attempted a comparison of this algorithm with our method.

C. A special iterative method for drops in close approach and extreme viscosity ratio

A traditional method to solve boundary-integral equations (1) is that of successive substitutions of the previous iterate $\mathbf{u}^{(\nu+1)}$ into the right-hand side (the so-called “simple interactions”). It is well known^{23,43} that all the spectrum of the double-layer integral operator (i.e., the values of κ for

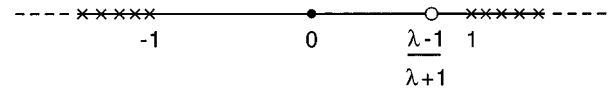


FIG. 6. The spectrum (crosses) of the double-layer integral operator.

which the homogeneous problem (1) has non-zero solutions) lies on the real axis outside $(-1, 1)$ (crosses in Fig. 6); marginal values -1 and $+1$ belong to the spectrum (Pozrikidis²³). This leads to the convergence of successive substitutions for all meaningful $\lambda > 0$ with the progression exponent $|\kappa|$. The deflation (9) removes $\kappa = \pm 1$ from the spectrum, thus accelerating the convergence, so that the progression exponent becomes $|\kappa|/r_1$, where r_1 is the minimum modulus of the spectrum points κ lying outside $[-1, +1]$. This deflation generally leads to successful iterations by successive substitutions, even in the limiting cases $\lambda \gg 1$ or $\lambda = 0$ (Loewenberg and Hinch⁴¹).

A non-trivial situation occurs when the surfaces are close. In this case, the spectrum of the boundary-integral operator (1) becomes nearly continuous, i.e., the density of crosses in Fig. 6 tends to infinity (similar to how the exact solution of two-sphere boundary-value problems in bispherical coordinates with a discrete spectrum is replaced by Fourier–Bessel integrals when the spheres touch). Obviously, since $r_1 \approx 1$ for close surfaces, the deflation (9) leaves the successive substitutions very poorly convergent if $\lambda \ll 1$ or $\lambda \gg 1$. Indeed, for $\lambda = 10^{-3}$ and the gap thickness on the order of 0.015 of the drop radii, we observed the number of iterations for successive substitutions exceeding 100, which is unacceptable for dynamical simulations. This deficiency makes very difficult, for example, numerical studies of deformable drop/bubble coalescence with $\lambda \ll 1$, and boundary-integral calculations for solid particles ($\lambda = \infty$) in close approach. In an analogous mathematical situation, with heat transfer through a very dense (or closely packed) system of highly conducting spheres, Zinchenko^{42,45} found a significant advantage of conjugate gradient iterations, with up to an order-of-magnitude saving in the number of iterations. However, (1) or (9) is not a self-adjoint system, and so conjugate gradient iterations cannot lead to an improvement. We have instead found that considerable progress can be made using a certain combination of *biconjugate* gradient iterations and simple iterations, as described below. Although the biconjugate gradient method of Lanczos⁴⁶ is a known abstract procedure for solving linear non-symmetric systems, we are not aware of any previous applications of this method to boundary-integral problems.

Having written the boundary-element discretization to (9) in a general matrix form

$$\mathbf{A}\mathbf{w} = \mathbf{b}, \quad (24)$$

one can generate a series of vectors $\mathbf{r}_k$, $\overline{\mathbf{r}}_k$, $\mathbf{p}_k$, $\overline{\mathbf{p}}_k$ and $\mathbf{w}_k$ as follows:⁴⁶

$$\mathbf{r}_{k+1} = \mathbf{r}_k - \alpha_k \mathbf{A} \mathbf{p}_k, \quad (25)$$

$$\overline{\mathbf{r}}_{k+1} = \overline{\mathbf{r}}_k - \alpha_k \mathbf{A}^T \overline{\mathbf{p}}_k, \quad (26)$$

$$\mathbf{w}_{k+1} = \mathbf{w}_k + \alpha_k \mathbf{p}_k, \quad (27)$$

where

$$\alpha_k = (\mathbf{r}_k, \overline{\mathbf{r}}_k) / (\mathbf{A} \mathbf{p}_k, \overline{\mathbf{p}}_k), \quad (28)$$

and

$$\mathbf{p}_{k+1} = \mathbf{r}_{k+1} + \beta_k \mathbf{p}_k, \quad (29)$$

$$\overline{\mathbf{p}}_{k+1} = \overline{\mathbf{r}}_{k+1} + \beta_k \overline{\mathbf{p}}_k, \quad (30)$$

where

$$\beta_k = (\mathbf{r}_{k+1}, \overline{\mathbf{r}}_{k+1}) / (\mathbf{r}_k, \overline{\mathbf{r}}_k). \quad (31)$$

Here $\mathbf{A}^T$ is the transpose of $\mathbf{A}$, $\mathbf{r}_0 = \mathbf{b} - \mathbf{A} \mathbf{w}_0$ is the residual of the initial approximation $\mathbf{w}_0$, $\mathbf{w}_k$ is the k th iterate of the solution, $\mathbf{p}_0 = \mathbf{r}_0$, and $\overline{\mathbf{p}}_0 = \overline{\mathbf{r}}_0$ is an arbitrary vector; in the present context, parentheses denote the usual L_2 -scalar product on $S_1 + S_2$, discretized by the trapezoidal rule. Since $\mathbf{A}$ is non-symmetric, the denominators in (28) and (31) can be of either sign. Negative $(\mathbf{r}_k, \overline{\mathbf{r}}_k)$ were found not to break the algorithm (in double precision calculations), but considerably hamper the convergence; the frequency of this event depends on the choice of an arbitrary initial vector $\mathbf{r}_0$. In the exact formulation, it is easy to prove that if $\mathbf{p}$ and $\overline{\mathbf{p}}$ are related so that $\overline{\mathbf{p}}$ represents the boundary value of the stress vector for the Stokes flows inside the drops with the velocities $\mathbf{p}$ on S_1 and S_2 , then $\mathbf{A}^T \overline{\mathbf{p}}$ is the boundary stress vector for the Stokes flows inside the drops with the velocity $\mathbf{A} \mathbf{p}$ on S_α . In this case, $(\mathbf{p}, \overline{\mathbf{p}}) > 0$ represents the rate of energy dissipation inside the drops (to within a factor), and $(\mathbf{A} \mathbf{p}, \overline{\mathbf{p}}) > 0$ follows from the fact that all eigenvalues of $\mathbf{A}$ are positive. Thus, if $\overline{\mathbf{r}}_0$ and $\mathbf{r}_0$ are related as the corresponding stress and velocity, then, in principle, (25)–(31) becomes a numerically stable process with positive denominators (28) and (31), similar to conjugate gradient iterations. However, due to finite-element discretization errors, this property of $\mathbf{A}$ is quickly distorted, and, in practice, we have found $\overline{\mathbf{r}}_0 = \mathbf{r}_0$ to be a much better *ad hoc* choice. With this choice, negative $(\mathbf{r}_k, \overline{\mathbf{r}}_k)$ occur very rarely and only in worst cases, when the convergence is not reached in a few iterations. We have found, as the best strategy, to monitor $(\mathbf{r}_{k+1}, \overline{\mathbf{r}}_{k+1})$ and, when it becomes negative, return to the previous iterate $\mathbf{w}_k$ and make one simple iteration $\tilde{\mathbf{w}} = \mathbf{r}_k + \mathbf{w}_k$, and then restart the whole biconjugate gradient process (25)–(31) from the obtained approximation $\tilde{\mathbf{w}} = \mathbf{w}_0$. The ultimate convergence is monitored by the magnitude of the residual $\mathbf{r}_k = \mathbf{b} - \mathbf{A} \mathbf{w}_k$. This combined procedure, in particular, takes advantage of the fact that the convergence of successive substitutions in boundary-integral calculations can be fast at first iterations, even for extreme λ and close approach, before it slows down dramatically, and of asymptotically much faster convergence for biconjugate gradient iterations at $k \rightarrow \infty$. By analogy with the properties of conjugate gradient iterations,⁴⁷ the progression exponent for the convergence of this method is expected to be $|\kappa|(1 - O(\sqrt{\epsilon}))$, compared to $\approx |\kappa|(1 - \epsilon)$ for successive substitutions, where $\epsilon = r_1 - 1 \ll 1$. For the success of this method, it is essential that $\mathbf{A}^T$ is taken as the exact adjoint to the finite boundary-element discretization $\mathbf{A}$. Both $\mathbf{A} \mathbf{p}_k$ and $\mathbf{A}^T \overline{\mathbf{p}}_k$ are calculated in the same loop over all pairs ($i < j$) of collocation points, and so iterations (25)–(31) are

only 50% more time consuming than usual successive substitutions. For $0.2 < \lambda < 5$, successive substitutions are competitive for all separations. However, in more extreme cases, the new iterative approach can save as much as an order-of-magnitude in the number of iterations, which will be demonstrated in Sec. IV for the bubble–bubble coalescence problem.

D. A novel mesh stabilization algorithm

Having solved the boundary-integral Eq. (9), one can advect the collocation points either with the interfacial velocity $\mathbf{u}$, given by Eq. (10), or with the normal velocity $(\mathbf{u} \cdot \mathbf{n}) \mathbf{n}$ to update the drop shapes. A familiar difficulty with both strategies is that the internode distances become highly irregular, invalidating the mesh after a short simulation time. Loewenberg and Hinch,⁴¹ who discuss this problem for equal drops freely suspended in a shear flow, suggest that the collocation points be convected with the velocity

$$\mathbf{v} = \mathbf{u} + \mathbf{\Pi} \quad (32)$$

as a remedy for this problem. The artificial non-dimensional tangential field $\mathbf{\Pi}$ at a vertex $\mathbf{x}^i$ was constructed as⁴¹

$$\mathbf{\Pi}^i = \psi \frac{N_\Delta^{3/2}}{300} \frac{1}{1 + \lambda} (\mathbf{I} - \mathbf{n}_i \mathbf{n}_i) \sum_j \left[1 + \frac{4|\mathbf{x}^j - \mathbf{x}^i|}{h(\mathbf{x}^j)} + |2k_j|^{3/2} \right] \Delta S_j (\mathbf{x}^j - \mathbf{x}^i), \quad (33)$$

using a (the non-deformed drop radius) and σ/μ_e as the characteristic length and velocity. In (33), ψ is the “grid tension” coefficient, k_j is the mean curvature at $\mathbf{x}^j$, $h(\mathbf{x}_j)$ is the distance (node-to-node minimum) from $\mathbf{x}^j$ to another drop surface, and ΔS_j is the portion of the surface area associated with $\mathbf{x}^j$; the summation is over all vertices directly connected to $\mathbf{x}^i$. In Ref. 41, relation (33) is given with $\psi = 1$, and our mean curvatures k_j differ in definition from those in Ref. 41 by a factor of 1/2, thus changing the appearance of this relation. The orders of magnitude of the coefficients in (33) were chosen to give the tangential field enough time to move nodes into the near-contact regions between drops and/or to the regions of high curvature,⁴¹ or approximately equalize the internode distances in cases of well-separated, nearly spherical surfaces.

While the above approach successfully controls the mesh for drops freely suspended in a shear flow,⁴¹ we have found that this approach, even with an optimum choice of the grid tension ψ , is inefficient for drops in gravity-induced motion. Namely, the stable time steps were found to be extremely small even for crude meshes, and prohibitively small for high resolutions, especially when $\lambda \ll 1$ (see Sec. V). A substantial difference between the two problems is that, for drops suspended in a shear flow, the fluid mostly circulates along the interfaces, and so advecting the mesh points with the interfacial velocity only does allow one to obtain a solution for a relatively long time, before it breaks down. In contrast, in gravity-induced motion, there is a very strong tendency for the collocation points to accumulate in the rear stagnation zones, leading to quick mesh degradation. For an

isolated spherical drop moving with a velocity $\mathbf{V}$, the normal velocity in the reference system associated with the drop, $(\mathbf{V} \cdot \mathbf{n})\mathbf{n} - \mathbf{V}$, has the same $\sim \sin\theta$ angular dependence as does the relative fluid velocity $\mathbf{u} - \mathbf{V}$; this helps to explain why using the normal velocity, instead of the interfacial velocity, does not solve the problem of mesh stabilization.

Our algorithm for mesh stabilization, suitable for any kind of flow, also uses an additional tangential field, but the construction of this field is *global* and based on a completely different idea. Note first that no tangential field can fully control the triangulation of a 3D surface. Indeed, there are about $1.5 N_\Delta$ mesh edges, but only about N_Δ free parameters in the additional tangential field; for this reason, one cannot set all (or almost all) internode distances to have prescribed values, or to be equal. Instead, we find an additional tangential field to minimize, on the average, the rates of change of the distances between the neighboring nodes. This optimization problem is solved separately for each surface. At any instant of time, the vertex velocities $\mathbf{V}_i = d\mathbf{x}^i/dt$ to be used in the shape updates are required to provide a minimum to the function

$$F(\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_M) = \sum_{(i,j)} \left[\frac{d}{dt} \mathbf{x}_{ij}^2 \right]^2 = 4 \sum_{(i,j)} [\mathbf{x}_{ij} \cdot (\mathbf{V}_j - \mathbf{V}_i)]^2 \quad (34)$$

[where M is the number of vertices on S_α , $\mathbf{x}_{ij} = \mathbf{x}^j - \mathbf{x}^i$, and the summation in (34) is over all triangle edges (i,j) on S_α] under the constraints

$$\mathbf{V}_i \cdot \mathbf{n}_i = q_i, \quad (35)$$

with the normal velocities $q_i = \mathbf{u}_i \cdot \mathbf{n}_i$ given by the solution of the boundary-integral Eqs. (9) and (10). In particular, for a non-deformed drop, problems (34)–(35) have a trivial solution $F=0$ and the internode distances do not change at all, which is a highly favorable property. For a deformable surface, problems (34)–(35) of conditional minimization are conveniently solved by gradient iterative methods. The solution $\mathbf{V}_i'$ from the preceding time step gives the initial approximation

$$\mathbf{V}_i^{(0)} = q_i \mathbf{n}_i + \mathbf{V}_i' - (\mathbf{V}_i' \cdot \mathbf{n}_i) \mathbf{n}_i. \quad (36)$$

The first iteration is made by the steepest descent method, moving along the surface in the direction opposite to $\partial F / \partial \mathbf{V}_i$:

$$\mathbf{V}_i^{(1)} = \mathbf{V}_i^{(0)} + \xi \mathbf{f}_i^{(0)}, \quad (37)$$

where

$$\mathbf{f}_i^{(v)} = (\mathbf{I} - \mathbf{n}_i \mathbf{n}_i) \sum_j [\mathbf{x}_{ij} \cdot (\mathbf{V}_j^{(v)} - \mathbf{V}_i^{(v)})] \mathbf{x}_{ij} = -\frac{1}{8} \left[\frac{\partial F(\mathbf{V}_1^{(v)}, \dots, \mathbf{V}_M^{(v)})}{\partial \mathbf{V}_i^{(v)}} \right], \quad (38)$$

the summation is over all neighbors j of a vertex i , and $\parallel$ means the projection on the tangential plane. The parameter

ξ is determined upon substituting (37) into (34) and minimizing $F(\mathbf{V}_1^{(1)}, \dots, \mathbf{V}_M^{(1)})$. The subsequent iterations correspond to the conjugate gradient method

$$\mathbf{V}_i^{(v+1)} = \mathbf{V}_i^{(v)} + \xi \mathbf{f}_i^{(v)} + \eta (\mathbf{V}_i^{(v)} - \mathbf{V}_i^{(v-1)}), \quad (39)$$

where ξ and η are determined upon substituting (39) into (34) and minimizing $F(\mathbf{V}_1^{(v+1)}, \dots, \mathbf{V}_M^{(v+1)})$. Obviously, all $\mathbf{V}_i^{(v)}$ satisfy the constraints (35). The process terminates when F stabilizes, $F^v - F^{v+1} < \epsilon F^{v+1}$. We used typically $\epsilon \sim 10^{-6}$ and, accordingly, 10–50 iterations, depending mostly on N_Δ . The quality of mesh stabilization was very insensitive to smaller ϵ ; on the other hand, several iterations are usually insufficient. In the optimal form, this, $O(N_\Delta)$ -part takes only a small portion of the total calculation time, even for $\lambda = 1$. The steepest descent method (37) could be used for all iterations, but it makes the convergence much slower.

A principal difference of our mesh stabilization method from the grid tension method⁴¹ (32)–(33) is that the latter seeks to ‘actively’ affect the distribution of mesh points (producing some ‘tension’ between the nodes), while our method is ‘passive,’ seeking only to preserve the current internode distances. Mathematically, (33) is an explicit local expression, while in our method all the velocities $\mathbf{V}_i$ are interrelated through a sparse matrix of mesh connectors. Also, note that our method uses only the normal velocity (35) of the fluid motion and is, therefore, a general algorithm for deformable surfaces.

Our ‘passive’ mesh stabilization algorithm with fixed connections can be generalized and made adaptive to near-contact zones (and, more difficult, to higher curvature zones); however, this most recent development was not included in the calculations of the present paper. The idea of mesh adaptation may be not attractive for numerical simulations of rheology and sedimentation of highly concentrated emulsions, because of multiple near-contact zones for each drop. So, our algorithm in the present form, with its softer stability constraints (much softer in case of gravity-induced motion), seems very suitable for these types of long-time simulations, especially sedimentation, if the drop shapes remain reasonably compact.

There exists another, completely different approach to the mesh control (Unverdi and Tryggvason⁴²), based on dynamical restructuring. To be viable for gravity-induced motion, however, this method should not only include reconnecting points, but also generating new and merging some old nodes,⁴² which makes this approach clearly more complicated, compared to our algorithm with fixed topology; the comparison between the efficiencies of the two methods remains an open question.

E. Numerical tests for shear-induced motion

In the present paper, we mostly focus on the gravity-induced motion of drops. However, an example for two equal drops freely suspended in a steady simple shear $(\gamma x_2, 0, 0)$ is considered to check the universality of our approach. The simulation presented in Fig. 7 was performed for $\lambda = 0.2$, $Ca = \mu_e \gamma a / \sigma = 0.35$, and 1280 triangles per drop,

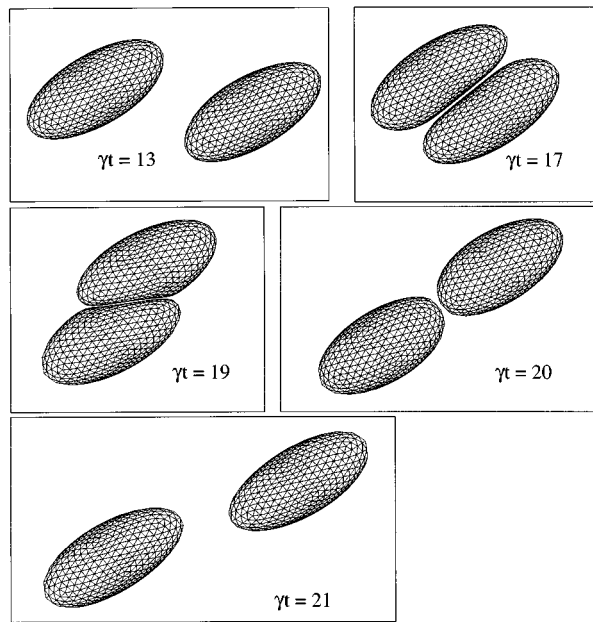


FIG. 7. Simulation of the relative motion of two equal drops freely suspended in a steady simple shear ($\gamma x_2, 0, 0$) using the present code with 1280 triangles per drop; $\mu_e \gamma a / \sigma = 0.35$, $\lambda = 0.2$, initial center-to-center offsets $\Delta x_1 = 10a$, $\Delta x_2 = 0.5a$, and $\Delta x_3 = 0$.

using our methods for the mean curvature and normal vector calculation (Sec. III B) and for the mesh stabilization (Sec. III D). The drops started from the spherical shape far upstream, with the initial offsets of the centers $\Delta x_1 = 10a$, $\Delta x_2 = 0.5a$, and $\Delta x_3 = 0$. As Fig. 7 shows, our mesh stabilization method maintains approximately uniform grids throughout the whole simulation. Except for the short initial period $\gamma t \sim 1.6$ of isolated drop relaxation from the spherical to the “quasi-steady” elongated shape, where a smaller step is necessary to maintain the accuracy, we have been able to integrate with the time step $\Delta t_o = 0.1 \gamma^{-1}$. When the drops come close, the time step should be reduced again to maintain accuracy, and we used the refinement rule $\Delta t = \Delta t_o (h_{min}/3a)^{1/2}$ in the “near zone” $h_{min} < 3a$, where h_{min} is the surface clearance (calculated as described in Sec. IV); a similar refinement rule was used in most of our calculations for gravity-induced motion (Sec. IV), although it makes the time step for close surfaces well below the stability limit and may be not optimal. To iterate (9) to the absolute residual 10^{-3} , typically from one to several iterations were required, and each half of the time step in the second order Runge–Kutta method took, on the average, 5 s on an IBM AIX RISC/6000 workstation. The accuracy of this simulation is confirmed in Figs. 8(a) and 8(b) by a similar run with 2420 triangles per drop and $\Delta t_o = 0.05 \gamma^{-1}$ (solid lines are for $N_\Delta = 1280$ and dark squares are for $N_\Delta = 2420$), and very good agreement between the results for the two meshes is seen both for the drop length L as a function of total strain γt and for the minimum surface clearance h_{min} as a function of the β -angle between the line-of-centers and the ambient flow direction. An additional, very extensive run with $N_\Delta = 4500$ and $\Delta t_o = 0.045 \gamma^{-1}$ was also generated by our code [dotted lines in Figs. 8(a) and 8(b)],

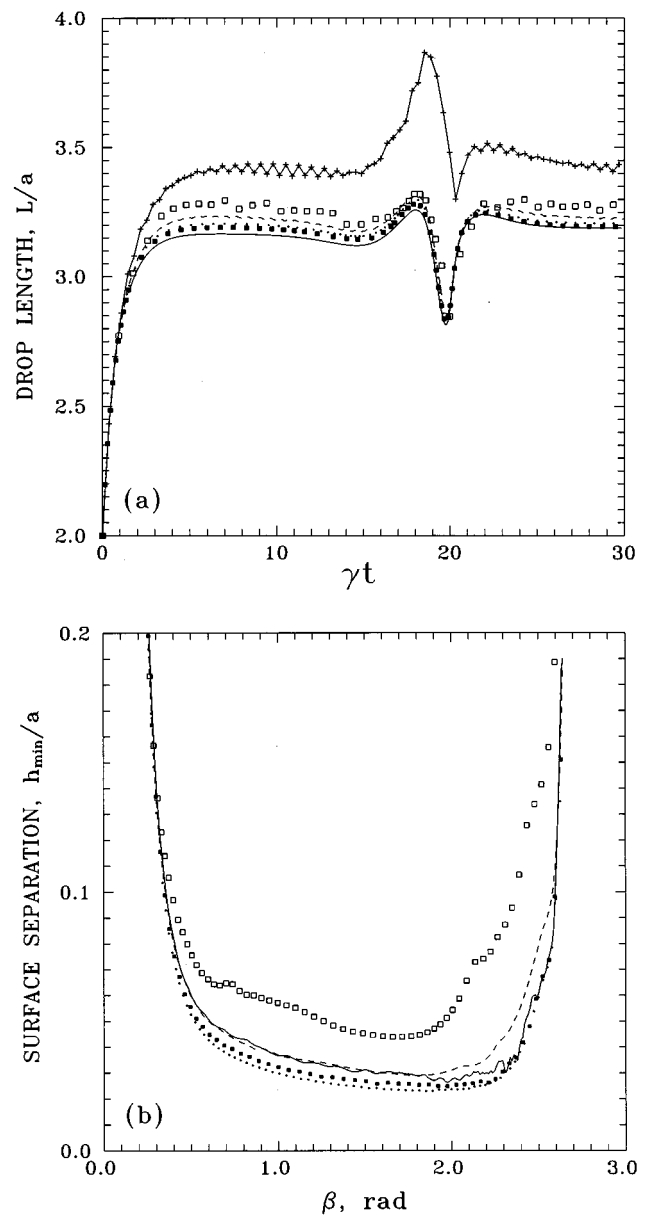


FIG. 8. Non-dimensional (a) drop length L/a vs strain γt and (b) surface separation h_{min}/a vs β -angle between the line-of-centers and the direction of the ambient flow (β increases as time proceeds) for the drop motion shown in Fig. 7. Solid lines, dots and dark squares are for the simulations using both the present curvature and mesh stabilization algorithms, with 1280 (solid lines), 2420 (dark squares) and 4500 (dots) triangle elements per drop. Open squares and dashed lines are for the simulations using the present curvature algorithm and the mesh stabilization (33) with $\psi = 0.2$, and 320 (open squares) and 1280 (dashed lines) elements per drop. The plus symbols in Fig. 12a are the calculations for 320 elements per drop by a previous code (Ref. 41). Drop length was evaluated as a maximum of inter-node distances; surface separation is calculated as described in Sec. IV.

but the results for the surface separation were found to be very close to those with $N_\Delta = 2420$, and the drop length L is almost indistinguishable from that with $N_\Delta = 2420$.

When we switched to the mesh stabilization method described by (32)–(33) for the same problem, the values of ψ near 0.2 were found to provide the most acceptable stability constraints for the whole simulation [if ΔS_i is defined by (19)]. With $\psi = 0.2$, it was possible to use the time step

choice⁴¹ $\Delta t = KCa\gamma^{-1}\Delta x/a$, where Δx is the minimum distance between the nodes, and a chosen value of the constant $K=0.5$; appreciably higher Δt could not be used because of the instability. Using this mesh stabilization method, but the present paraboloid curvature method, two more simulations were generated, with $N_\Delta=320$ [open squares in Figs. 8(a) and 8(b)] and $N_\Delta=1280$ (dashed lines). The simulation with $N_\Delta=320$ is reasonably accurate for the drop length [Fig. 8(a)], but, as expected for low resolutions, is seriously in error in describing the delicate near-contact motion [Fig. 8(b)]. The simulation with $N_\Delta=1280$ using the adaptive mesh stabilization method (32)–(33) has a similar overall accuracy for the drop length calculation but becomes less accurate in describing the latter stages of the near-contact motion, compared to the simulation using our non-adaptive mesh stabilization method and the same N_Δ [cf. dashed and solid lines with the most accurate calculations represented by dots and dark squares in Figs. 8(a) and 8(b)]. We did not control how successfully the method (32)–(33) adapts the grid points to near-contact and/or high curvature zones; note only that the rate of the film drainage between close drops is not solely determined by local conditions, and so the mesh condensation in certain zones may have an overall negative effect, which is observed in Fig. 8(b) near the end of the near-contact motion. In addition, the stability limitations for the method (32)–(33) are much tighter: with $N_\Delta=1280$, our mesh stabilization method uses 6–7 times larger Δt for well-separated drops and 3.5 times larger Δt on the average for the whole simulation in Fig. 8, resulting in a more than two times faster calculation.

Also given in Fig. 8(a) (plus symbols) are calculations with $N_\Delta=320$ using a previous code⁴¹ for the same problem (these results were kindly provided by Professor M. Loewenberg). The curvature and normal vector calculations in this code are based on contour integration. The open squares, which represent our calculations with $N_\Delta=320$ using the same mesh stabilization method but the present curvature algorithm are much closer to the converged results (dark squares with $N_\Delta=2420$).

Although the accuracy of different characteristics associated with drop motion correlates with the accuracy of the curvature calculation [as the analysis in Fig. 8(a) shows], this relationship may be complex. As an example, Fig. 9 gives the maximum drop length L_{max}/a calculated by the present code (using paraboloid fitting for the curvature and the passive mesh stabilization technique) as a function of N_Δ^{-1} . A unified procedure of fragmenting each drop surface into $N_\Delta=20n^2$ mesh triangles (see Sec. III A) was used in these calculations; the parameter Δt_o determining the time step (see above) was set to $0.085(N_\Delta/1280)^{-1/2}$. The results in Fig. 9 appear to converge approximately linearly in N_Δ^{-1} for $N_\Delta \geq 720$, although we can only prove our local curvatures to be $O(N_\Delta^{-1/2})$ accurate at $N_\Delta \rightarrow \infty$. We tried to perform a similar convergence analysis for the center-to-center offset Δx_2^∞ in the velocity-gradient direction far downstream, but the latter characteristic, when calculated by the present code, was found to be very insensitive to the number of boundary elements; namely, for all the calculations in Fig. 9, the corresponding values of Δx_2^∞ (all evaluated at the

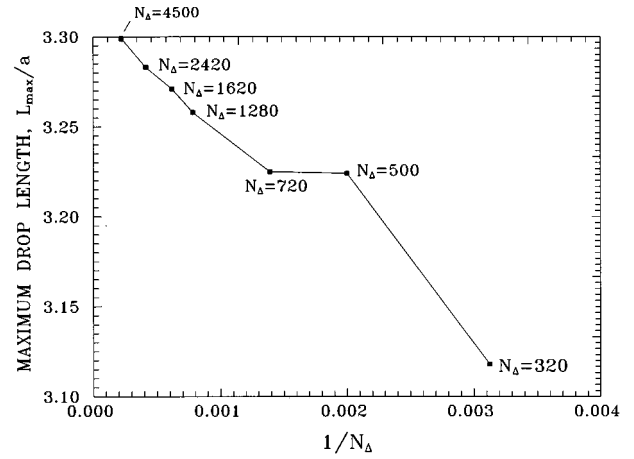


FIG. 9. The maximum non-dimensional drop length L_{max}/a calculated by the present code for a different number N_Δ of triangle elements per drop. The physical parameters are the same as for the simulations shown in Figs. 7–8.

same downstream offset $\Delta x_1 \approx 17a$) lie within tight limits 1.466–1.487 a (recall that $\Delta x_2 = 0.5a$ at the initial upstream offset $\Delta x_1 = 10a$, and so a net displacement occurs due to deformation), which makes an accurate convergence analysis in the latter case more difficult and less meaningful.

F. The removal of the curvature from the boundary-integral formulation

Interestingly, the mean curvature calculation, capable of causing inaccuracies, can be removed altogether from the boundary-integral formulation. Using the differential form of (21), one can integrate the curvature term in (2) by parts over the whole surface S_α and arrive at the following relation (see the Appendix):

$$2 \int_{S_\alpha} k(\mathbf{x}) \mathbf{n}(\mathbf{x}) \cdot \mathbf{G}^{(\prime)}(\mathbf{x}-\mathbf{y}) dS_x = \frac{1}{8\pi} \int_{S_\alpha} \frac{r_\alpha}{r^3} \left\{ 1 - \frac{3[\mathbf{r} \cdot \mathbf{n}(\mathbf{x})]^2}{r^2} \right\} dS_x; \quad (40)$$

if $\mathbf{y} \in S_\alpha$, the right-hand side (40) is the principal-value integral. Thus, the new formulation requires only the normal vectors $\mathbf{n}(\mathbf{x})$, which are usually much less sensitive than the curvature to discretization errors. The difficulty, however, is that the new form (40) is more singular when $\mathbf{y} \in S_\alpha$ (or $\mathbf{y}$ close to S_α), and a proper regularization is non-trivial. When $\lambda = 1$, so that only the normal component of the vector (40) is essential for boundary-integral calculations, further progress can be made by transforming (40) to

$$2 \int_{S_\alpha} k(\mathbf{x}) \mathbf{n}(\mathbf{x}) \cdot \mathbf{G}^{(\prime)}(\mathbf{x}-\mathbf{y}) n_\alpha(\mathbf{y}) dS_x = \frac{1}{8\pi} \int_{S_\alpha} \frac{\mathbf{r} \cdot [\mathbf{n}(\mathbf{x}) + \mathbf{n}(\mathbf{y})]}{r^3} \times \left\{ 1 - \frac{3[\mathbf{r} \cdot \mathbf{n}(\mathbf{x})][\mathbf{r} \cdot \mathbf{n}(\mathbf{y})]}{r^2} \right\} dS_x \quad (41)$$

(see the Appendix). Unlike in (40), the integrand in the right-hand side of (41) is non-singular, since $\mathbf{r} \cdot [\mathbf{n}(\mathbf{x}) + \mathbf{n}(\mathbf{y})] = O(r^3)$ for $\mathbf{y} \in S_\alpha$ and $\mathbf{x} \rightarrow \mathbf{y}$, and so the trapezoidal rule can be used. The transformation of (40) to (41) also has some advantage for closely interacting drops since, if $\mathbf{x}$ and $\mathbf{y}$ lie in the gap region on different surfaces, then the normals $\mathbf{n}(\mathbf{x})$ and $\mathbf{n}(\mathbf{y})$ approximately counterbalance, smoothing the integrand. Unfortunately, we have found this improvement insufficient, and numerical solutions using (41) to break down for drops in close approach. For well-separated surfaces, when this problem does not occur, the new, curvatureless form (41) gives considerably more accurate results for the same triangulation than does the conventional form, which will be demonstrated by the analysis of the drop breakup (Sec. IV). Further work is needed, however, to make the identity (40) useful for drops in close approach and/or with $\lambda \neq 1$.

IV. NUMERICAL RESULTS FOR BUOYANCY-DRIVEN MOTION

In the numerical results presented herein for two deformable drops of the same fluid settling under gravity through an immiscible fluid, the characteristic lengthscale is the largest of the two radii, $a_2 > a_1$, for non-deformed drops; the characteristic velocity V and time are $|\rho' - \rho_e|ga_2^2/(8\pi\mu_e)$ and a_2/V , respectively. The results are a function of size ratio a_1/a_2 and viscosity ratio λ , initial horizontal $\Delta\tilde{x}_0 = \Delta x_0/a_2$ and vertical $\Delta\tilde{z}_0 = \Delta z_0/a_2$ offsets of drop centers (tilde indicates non-dimensional values), and the capillary number:

$$Ca = \frac{\mu_e |\mathbf{V}_1^\infty - \mathbf{V}_2^\infty|}{\sigma}, \quad (42)$$

with $\mathbf{V}_i^\infty$ being the settling velocities of isolated spherical drops. Due to $\Delta\tilde{z}_0 \gg 1$, spherical shapes were always chosen as the initial condition.

As the first test (Fig. 10), comparison was made between the exact solution for two spherical drops with $a_1/a_2 = 0.7$, $\lambda = 3$, $\Delta\tilde{x}_0 = 0.45$, $\Delta\tilde{z}_0 = 5$ and our boundary-integral solution for the same a_1/a_2 , λ , $\Delta\tilde{x}_0$, $\Delta\tilde{z}_0$ and $Ca = 0.0723$, when the effect of deformation is expected to be small. The relative center-to-center trajectories for non-deformed and deformed drops are represented by solid and dashed lines, respectively. Also shown are the drop positions (solid lines for non-deformed and dotted lines for deformed drops) at the final shown value of the β -angle between the centerline and the vertical. The exact bispherical coordinate solutions for the motion along (Haber *et al.*²) and normal (Zinchenko³) to the line-of-centers were used to calculate the relative trajectory for non-deformed drops. In the boundary-integral solution, $N_\Delta = 3840$ triangles per drop and the initial time step $\Delta\tilde{t}_0 = 0.01$ (close to the stability limit) were used; for $\tilde{h}_{min} < 3$ the time step was refined in the same manner as for shear flow (Sec. III D). Excellent agreement between the bispherical coordinate and boundary-integral solutions can be seen from Fig. 10, except for small separations when the deformation becomes important. Despite the simplicity of relative trajectories (Fig. 10), a $Ca \ll 1$ test is not a trivial

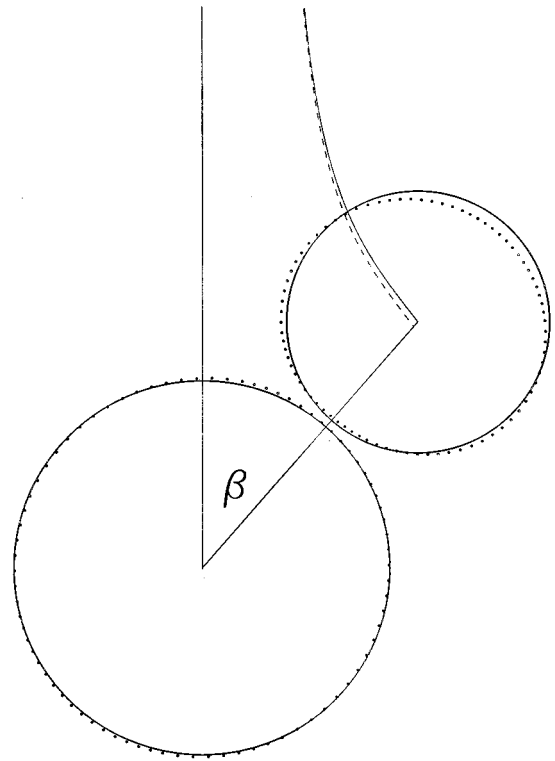


FIG. 10. Comparison between the exact bispherical coordinate solution (solid lines) for $a_1/a_2 = 0.7$, $\lambda = 3$, $\Delta\tilde{x}_0 = 0.45$, $\Delta\tilde{z}_0 = 5$, and the boundary-integral solution (dashed line for the relative center-to-center trajectory and dotted lines for the drop positions) with the same a_1/a_2 , λ , $\Delta\tilde{x}_0$, $\Delta\tilde{z}_0$, and $Ca = 0.0723$.

one: it can be seen from (1), (2) and (7) that the small $O(Ca)$ -correction to the spherical drop shapes in the boundary-integral solution has an $O(1)$ -effect on the relative trajectory for $Ca \rightarrow 0$.

In the example of Fig. 10, further motion of spherical drops leads to collision (because of the integrable singularity $\sim h_{min}^{-1/2}$ of hydrodynamic resistance for spherical drops^{4,8}) at $\beta = 68.7^\circ$; in contrast, the asymptotic theory of Yiantsios and Davis^{19,20} (although developed for axisymmetric approach) predicts that a small amount of deformation in the gap region prohibits physical contact between the surfaces (unless van der Waals attraction is included). For further applications of our code to coalescence phenomena, it is of particular interest to check the consistency of our boundary solution with this prediction and find the actual minimum separation that two slightly deformable drops can reach without van der Waals attraction. To reduce the numerical difficulties, we have studied this question for a somewhat higher Ca and $\lambda = 1$. Figure 11 presents the relative motion of two slightly deformable drops with $a_1/a_2 = 0.7$, $\lambda = 1$, $Ca = 0.1445$, $\Delta\tilde{x}_0 = 0.325$, $\Delta\tilde{z}_0 = 5.087$, $N_\Delta = 3840$ triangles per drop, and $\Delta\tilde{t}_0 = 0.0096$ (close to the stability limit). Our stabilization method of Sec. III D keeps the meshes almost uniform throughout the whole simulation, with the maximum-to-minimum mesh edge ratio r_{max}/r_{min} not exceeding 1.28 for both drops. Each half of the time step in the second-order

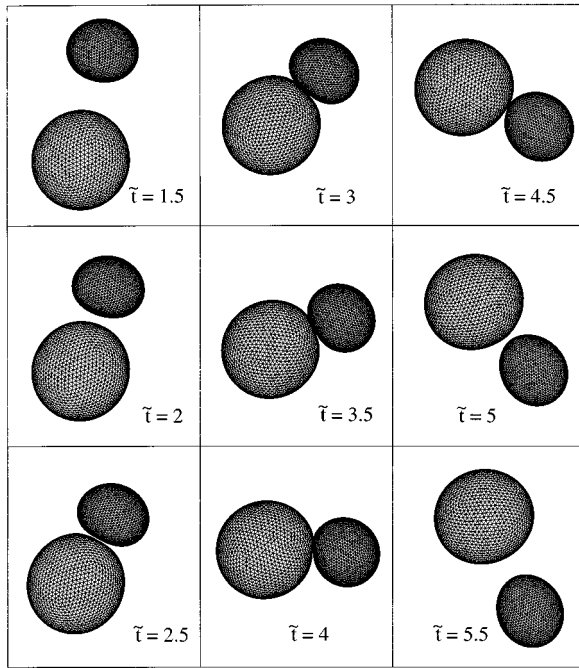


FIG. 11. Relative gravity-induced motion of two drops with $a_1/a_2=0.7$, $\lambda=1$, $Ca=0.1445$, $\Delta\tilde{x}_0=0.325$, and $\Delta\tilde{z}_0=5.087$ by the boundary-integral simulation with 3840 triangle elements per drop.

Runge–Kutta integration scheme took about 12 s of CPU time on an IBM AIX RISC/6000 workstation. Spherical drops in this case would collide at $\beta=29.3^\circ$. In contrast, the boundary-integral solution (Fig. 11) shows that the drops stay very close, but do not touch, and eventually separate. As in Sec. III E, the separation h_{min} between two triangulated surfaces was rigorously calculated as the minimum of node-to-node, node-to-face, node-to-edge, and edge-to-edge distances; it can be shown that for a generic case only these possibilities should be considered. Figure 12 presents $\tilde{h}_{min}(\beta)$ for the drop motion in “apparent contact.” As a convergence check, we have also performed very intensive calculations with $N_\Delta=15\,360$ triangles per drop and $\Delta\tilde{t}_0=0.0048$; the agreement between the two solutions (Fig. 12) is excellent, considering how small the surface clearance becomes, about $0.007a_2$. Hopefully, in future studies with additional non-hydrodynamic attractive forces, the present reliable numerical solution can be used to delineate the conditions leading to drop coalescence. Note, however, that the convergent calculations of this type become exceedingly difficult as Ca is decreased due to (i) very tight stability constraints $\Delta\tilde{t}_0 < O(CaN_\Delta^{-1/2})$ and (ii) the localization of stress in the small gap region; at the same time, $Ca \ll 1$ is a typical practical case for drops moving under low Reynolds number conditions. The idea of an adaptive mesh in the gap region, unfortunately, is less attractive than it seems because the film drainage rate is not solely determined by the local conditions. We noticed, in some calculations, an overall loss of accuracy for strongly non-uniform meshes; besides, the mesh condensation in the gap region makes the stability constraints much tighter. As an alternative to very difficult boundary-integral calculations for $Ca \rightarrow 0$, we have recently

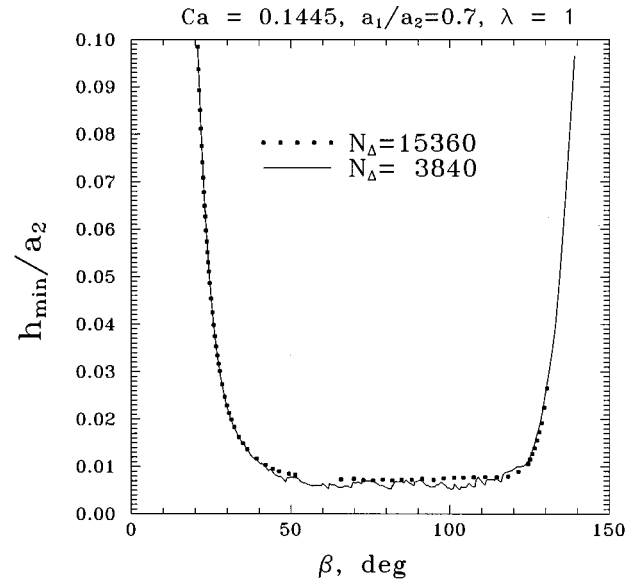


FIG. 12. Non-dimensional surface clearance, h_{min}/a_2 , vs β -angle between the line of surface centroids (12) and the vertical (β increases as time proceeds) for the simulation shown in Fig. 11 (solid line) and for a similar simulation using 15 360 triangle elements per drop (squares).

developed an asymptotic theory of coalescence of horizontally offset, slightly deformable drops in buoyancy-driven motion, based on matching the outer solution for spherical drops (valid on the drop lengthscale) with the solution of integro-differential thin film equations in the small gap region²¹. However, the range of validity of the asymptotic theory can be found only by the comparison with the boundary-integral solutions, and we have found the information in Fig. 12 very valuable for this purpose.²¹

Large drop deformations lead to a number of qualitatively new features. Of particular interest are the experiments of Manga and Stone^{33,40} for two air bubbles in a highly viscous corn syrup. They discovered, in particular, that a smaller bubble can be swept around the larger one and then sucked in, with eventual coalescence; the critical initial horizontal offset Δx_0 leading to the capture and coalescence can even exceed the Smoluchowski value of $a_1 + a_2$! To check if our code can successfully simulate this interesting phenomenon, we have chosen an extremely small viscosity ratio $\lambda=10^{-3}$, $Ca=0.723$, $a_1/a_2=0.7$ and initial offsets $\Delta\tilde{z}_0=10$ and $\Delta\tilde{x}_0=1.7$ (i.e., exactly the non-dimensional a_1+a_2). The simulation presented in Fig. 13 for 3840 triangles per drop, indeed, demonstrates that after the small drop is swept around the larger one, it becomes sucked into the dimple formed on the larger drop, in qualitative agreement with the experiments^{33,40} (performed at larger capillary numbers). Figure 13 also demonstrates that our mesh stabilization method (Sec. III D) maintains a well-controlled grid throughout the whole simulation, the mesh edge ratios r_{max}/r_{min} being within 1.29 for the larger and 2.19 for the smaller drop. The dynamics of the thin film drainage between the drops in this simulation was also considered by plotting the surface clearance $\tilde{h}_{min}(\beta)$ (Fig. 14). The calculations for 3840 and 2420 triangles per drop demonstrate

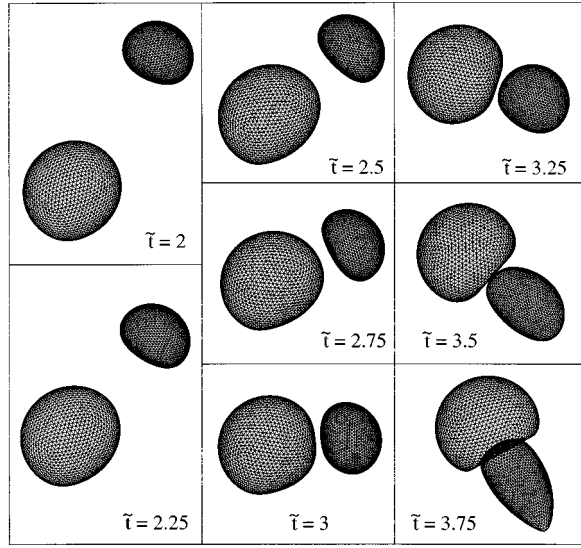


FIG. 13. Relative buoyancy-driven motion of two bubbles with $a_1/a_2=0.7$, $\lambda=10^{-3}$, $Ca=0.723$, $\Delta\tilde{x}_0=1.7$, and $\Delta\tilde{z}_0=10$ by the boundary-integral simulation using 3840 triangle elements per drop.

excellent convergence of $\tilde{h}_{min}(\beta)$ in the wide range of separations. The initial time step $\Delta\tilde{t}_0$ in the two simulations was 0.012 (about 70% of the stability limit for $N_\Delta=3840$). However, unlike for $\lambda=O(1)$, our usual empirical rule $\Delta\tilde{t}=\Delta\tilde{t}_0(\tilde{h}_{min}/3)^{1/2}$ for step refinement at $\tilde{h}_{min}<3$ was insufficient in this case of $\lambda\ll 1$ because of the fast film drainage rate, and, to accurately achieve the limit $\Delta\tilde{t}\rightarrow 0$ for each of the two curves in Fig. 14, much smaller steps $\Delta\tilde{t}$ were used for small gaps, setting $\Delta\tilde{t}_0=0.006$ for $0.03<\tilde{h}_{min}<0.3$ and $\Delta\tilde{t}_0=0.0003$ for $\tilde{h}_{min}<0.03$ in the above relation for $\Delta\tilde{t}$. The boundary-integral equation (9)

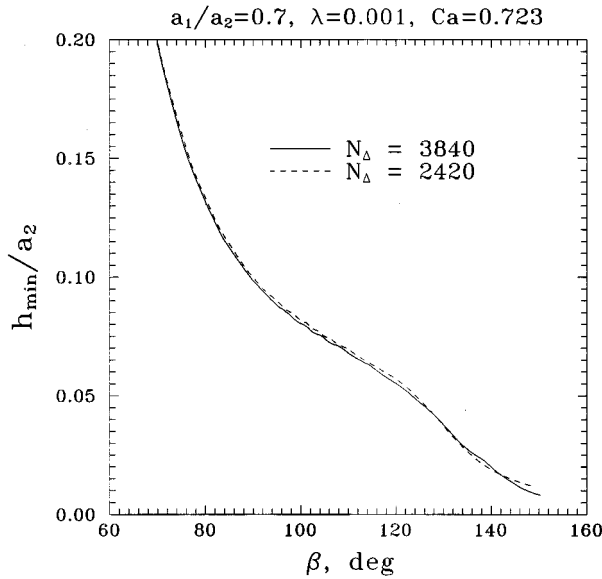


FIG. 14. Non-dimensional surface clearance, h_{min}/a_2 , vs β -angle for the simulation shown in Fig. 13 (solid line), and for a similar simulation using 2420 elements per drop.

was iterated to the absolute residual 10^{-3} (despite this seemingly crude criterion, the solutions were surprisingly insensitive to smaller tolerances; however, much higher tolerances quickly generate instability). To successfully achieve the small separations shown in Fig. 14, it was of particular importance to use the combined biconjugate gradient–simple iteration method of Sec. III C. In particular, in the simulation with $N_\Delta=2420$, the average number of iterations over 30 consecutive time steps was only 1.67 at $\tilde{t}=3.45(\tilde{h}_{min}\approx 0.049)$, 4.10 at $\tilde{t}=3.61(\tilde{h}_{min}\approx 0.02)$ and 5.64 at $\tilde{t}=3.66(\tilde{h}_{min}\approx 0.015)$. For comparison, when we switched temporarily to traditional successive substitutions, the corresponding average number of iterations was much higher, namely, 7.7, 22.2, and 57.1. Interestingly, the film drainage accelerates at a certain stage, before slowing dramatically (Fig. 14). Unfortunately, it was not possible to compare the film drainage rate at the latest stage with the existing asymptotic theories, which were developed for local deformations or plane parallel films and/or do not include the case $\lambda\ll 1$. Qualitatively, the film drainage rate should be similar to that for a spherical bubble of radius a entrapped in a liquid eccentric spherical drop of a close radius b with a free surface. Using the asymptotics⁴⁸ for the lubrication force $F\sim 2\pi\mu_e ab(b-a)^{-1}\ln(b/h_{min})V$, where V is the bubble velocity relative to the drop along the line-of-centers, and the approximation $b/a=1+O(h_{min}/b)$, we can roughly estimate the ultimate film drainage rate for the simulation in Fig. 13 as

$$d\tilde{h}_{min}/d\tilde{t}=O[\tilde{h}_{min}|\ln\tilde{h}_{min}|^{-1}].$$

Obviously, complete film drainage requires an infinite time (unless van der Waals attraction comes into play).

We have been unable to continue these simulations for $\tilde{t}>3.76$ because of the sharp tail forming on the smaller drop (Fig. 13); the occurrence of the tail was also observed experimentally by Manga and Stone.^{33,40} The presence of a sharp tail made our velocity iterations divergent for realistic values of N_Δ , presumably because the exact spectrum of the boundary-integral operator is continuous in this case. This problem, which may be the most serious obstacle for 3D boundary-integral simulations with very small λ , has not been addressed in the literature, to our knowledge. In the future, we will attempt different regularization schemes, but the simulations with nearly sharp tails are expected to nevertheless be difficult because of the numerical stiffness and slow convergence of iterations. Despite this complication, it can be predicted that the drops in Fig. 13 continue the motion as an “aggregate” to $\beta=\pi$ for an infinite time without separation. To show this, we have simulated an axisymmetrical situation, when the smaller drop is initially placed behind the larger one on the axis of symmetry ($\beta=\pi$) in close contact ($h_{min}=0.15a_2$), see Fig. 15(a). The initial shapes were taken as an oblate (larger drop) and prolate (smaller drop) spheroids (of the same volumes as non-deformed drops of radii a_1 and a_2) with the aspect ratio 1.1 each, i.e., much less deformed than at the end of the simulation in Fig. 13. The parameters $a_1/a_2=0.7$, $\lambda=10^{-3}$, $N_\Delta=3840$ are the same as in Fig. 13, but a smaller $Ca=0.506$ was taken to reduce the problems associated with the sharp tail formation.

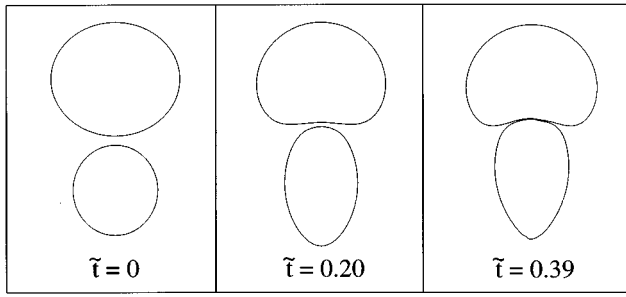


FIG. 15. The simulation of the axisymmetrical “sucking-in” phenomenon for two bubbles with $a_1/a_2=0.7$, $\lambda=10^{-3}$, and $Ca=0.506$.

If the drops were non-deformable spheres, they would separate because the larger leading drop moves faster. In contrast, the simulation in Fig. 15 shows that the smaller drop is being sucked into the larger one; the time dynamics of the surface clearance $h_{\min}(\tilde{t})/a_2$ are presented in Fig. 16. Interestingly, the shape of the aggregate at the end of the simulation qualitatively resembles the shape of a *single* low-viscosity drop caused by the development of the instability in buoyancy-driven motion (Fig. 4 of Pozrikidis³¹); we cannot exclude the possibility, however, that our aggregate attains a steady-state shape at $\tilde{t} \rightarrow \infty$. Since the “sucking-in” phenomenon is observed for $Ca=0.506$, the same is expected for higher capillary numbers, due to stronger deformation.

We did not try to extend the difficult calculations shown in Figs. 15 and 16, because this type of simulation should be done much more efficiently in the future by an axisymmetric code, rather than by our 3D algorithm. Efficient axisymmetric codes have been developed both for a single drop^{29–32} and for two drops.³³ For two drops, however, the axisymmetric calculations have been used so far in the case when the smaller drop is the leading one and acquires an oblate shape as the result of the interaction,³³ while our Fig. 15 suggests a

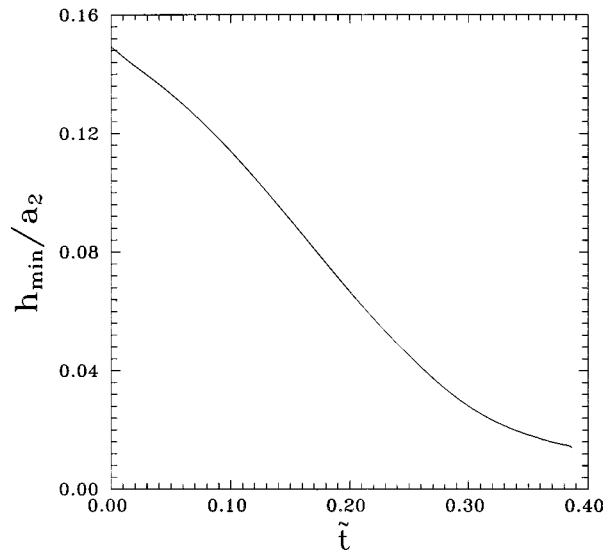


FIG. 16. The dynamics of the surface clearance, $\tilde{h}_{\min}(\tilde{t})$, for the simulation shown in Fig. 15.

different type of simulation, with the larger drop being the leading one. Hopefully, the present idea of using biconjugate gradient iterations (Sec. III C) will give additional strength to axisymmetric calculations for drops in close approach with extreme λ . Calculations similar to those in Fig. 15 can be used to find the boundary in the parameter space (Ca , a_1/a_2 , λ) demarcating the values for which the drop capture caused by deformations is possible. For each set of values (Ca , a_1/a_2 , λ) inside this boundary, 3D simulations similar to those in Fig. 13 can be used to find the capture efficiency as the maximum horizontal offset far upstream leading to capture by trials-and-errors. This interesting but computationally intensive task will be undertaken in our future work, as well as a comparison with the experiments.^{33,40} Note that in this formulation the capture efficiencies are independent of short-range van der Waals forces, although the latter are important in making the drops eventually coalesce.³³

To our knowledge, the calculations in Fig. 13 present the first adequate 3D boundary-integral simulation of bubble or drop capture enhanced by deformations. Manga and Stone, who discovered this remarkable phenomenon, presented 3D simulations for $\lambda=1$ and only the initial stage of mutual approach.⁴⁰ An extremely small λ , rather than unity, is expected to play a crucial role in this phenomenon for several reasons. First, very small λ decreases the drop tendency to break prematurely.^{49,50} Second, the far-field perturbation analysis of the force leading to drop alignment and capture reveals³³ that this force has a factor $c(\lambda)$ which is almost four times larger for $\lambda \rightarrow 0$ than for $\lambda=1$. Finally, very small λ drastically increases the rate of the thin film drainage (which is retarded by the drop viscosity when λ is large).⁸

Our last series of simulations, indeed, demonstrates that breakup is a typical result of hydrodynamic interaction for deformable drops with $\lambda=O(1)$ in gravity-induced motion, even for moderately large Ca . The simulation in Fig. 17 is for $\lambda=1$, $a_1/a_2=0.7$, $Ca=0.723$, initial offsets $\Delta\tilde{z}_0=5.087$ and $\Delta\tilde{x}_0=1$, and $N_\Delta=3840$ triangle elements on each drop; for $\tilde{t}=1.5$ and 4.5 , the contour comparison between this simulation (bold lines) and another simulation using 1280 triangles per drop (dashed lines) is also given. The simpler calculation with $N_\Delta=1280$ is quite acceptable in the early stages, but quickly loses accuracy after $\tilde{t}=4.5$ as the smaller drop stretches. A switch was made at $\tilde{t}=4.5$, when the drops are already well separated, to the curvatureless form (41), to generate the last two frames in Fig. 17, for $\tilde{t}=5$ and 5.5 . The accuracy of this new boundary-integral formulation (41) was assessed by repeating the whole simulation with $N_{\Delta_2}=3840$ and $N_{\Delta_1}=15360$ triangles on the smaller drop. The comparison of this, the most accurate simulation, with those using $N_{\Delta_1}=N_{\Delta_2}=3840$ [Figs. 18(a) and 18(b) and Fig. 19] shows that the new form (41) gives results much closer to the exact solution than does the conventional boundary-integral formulation, for this case of highly elongated drops. The initial time steps used are $\Delta\tilde{t}_0=0.048$ for $N_{\Delta_1}=N_{\Delta_2}=3840$, and $\Delta\tilde{t}_0=0.024$ for $N_{\Delta_2}=3840$, $N_{\Delta_1}=15360$, about 60% of the initial stability limit in both cases. For $h_{\min}/a_2 < 3$, the time step was refined in our usual way. However, due to the drop stretching,

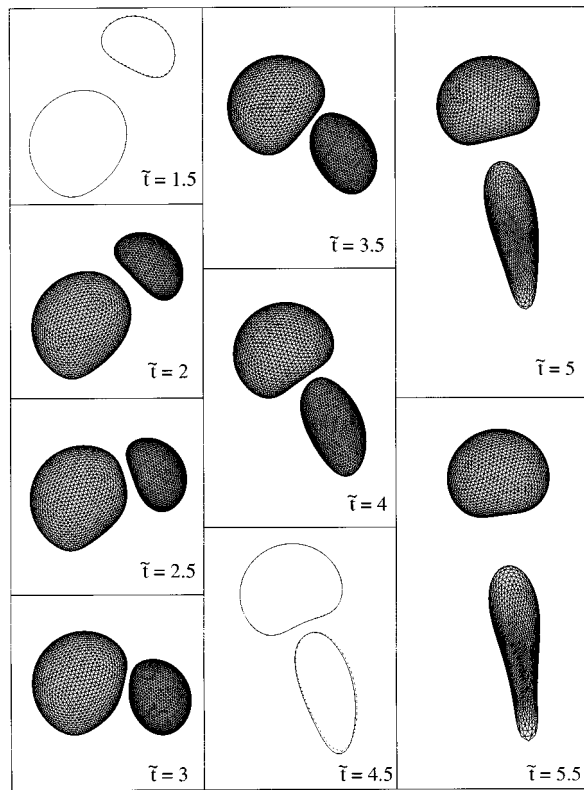


FIG. 17. Relative gravity-induced motion of two drops with $a_1/a_2=0.7$, $\lambda=1$, $Ca=0.723$, $\Delta\tilde{x}_0=1$, and $\Delta\tilde{z}_0=5.087$. The simulation shown for $\tilde{t}=2-4$ and $\tilde{t}=5, 5.5$ uses 3840 triangle elements per drop; for $\tilde{t}=1.5$ and 4.5, the comparison is given between this simulation (solid lines) and another one (dashed lines) with 1280 elements per drop.

the minimum mesh edge r_{min} may be quite small (see Fig. 17), and so an additional constraint $\Delta t \leq K r_{min}$ was imposed, with the same constant $K = \Delta\tilde{t}_0 / r_{min}^0$ as at the beginning of the simulation; the latter limitation, however, was important only for $\tilde{t} > 5.5$. The simulations took only about

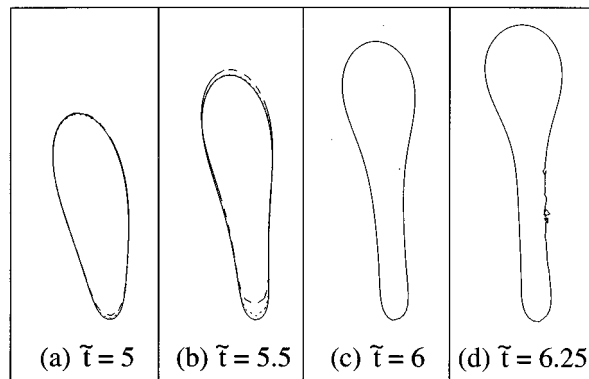


FIG. 18. The stretching of the smaller drop (the larger one remains compact and is not shown) in the simulations with $a_1/a_2=0.7$, $\lambda=1$, $Ca=0.723$, $\Delta\tilde{x}_0=1$, and $\Delta\tilde{z}_0=5.087$. The solid lines ($N_{\Delta_1}=15360$, $N_{\Delta_2}=3840$), and short-dashed lines ($N_{\Delta_1}=N_{\Delta_2}=3840$) are for the simulations using the curvatureless form (41) after $\tilde{t}=4.5$; the long-dashed lines ($N_{\Delta_1}=N_{\Delta_2}=3840$) are for the simulation with the conventional boundary integral formulation for all $\tilde{t}$.

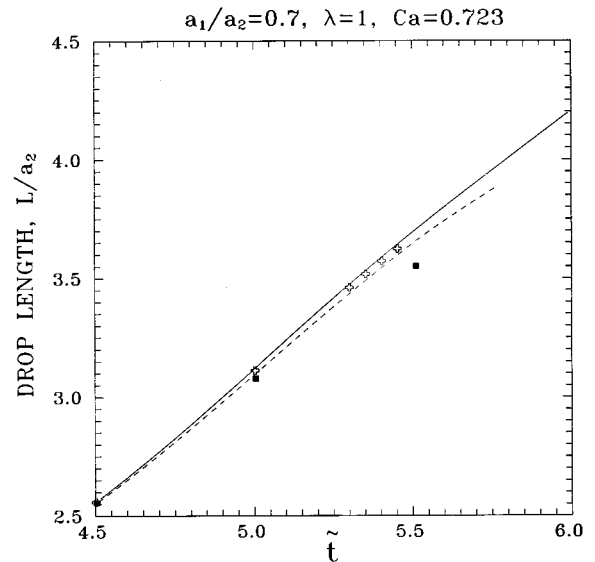


FIG. 19. The non-dimensional length L/a_2 of the smaller drop vs time, $\tilde{t}$, in the simulations with $a_1/a_2=0.7$, $\lambda=1$, $Ca=0.723$, $\Delta\tilde{x}_0=1$, and $\Delta\tilde{z}_0=5.087$. Squares ($N_{\Delta_1}=N_{\Delta_2}=3840$) and crosses ($N_{\Delta_1}=15360$, $N_{\Delta_2}=3840$) are for the simulations using the conventional boundary integral formulation; the solid line ($N_{\Delta_1}=15360$, $N_{\Delta_2}=3840$) and the dashed line ($N_{\Delta_1}=N_{\Delta_2}=3840$) are for the simulations using the curvatureless form (41) after $\tilde{t}=4.5$.

2.5 h each on the IBM AIX RISC/6000 workstation for $N_{\Delta_1}=N_{\Delta_2}=3840$, and 13 times longer, when $N_{\Delta_2}=3840$ and $N_{\Delta_1}=15360$. More expensive runs, with $\Delta\tilde{t}_0=0.024$ for $N_{\Delta_1}=N_{\Delta_2}=3840$, and $\Delta\tilde{t}_0=0.012$ for $N_{\Delta_2}=3840$ and $N_{\Delta_1}=15360$, were also generated, but these showed a negligible difference from the results in Figs. 17 and 18.

Although we could not assess the accuracy of our calculations with $N_{\Delta_1}=15360$ for $\tilde{t}=6$ and 6.25 shown in Figs. 18(c) and 18(d) (since we do not have reasonably good results with other meshes at long times), monitoring the volume v of the smaller drop (which was not forced to conserve in any way) gives an idea about the range of validity of the numerical solution. The volume fluctuation $|v-v_0|/v_0$ (where v_0 is the initial value) was within 0.5% for $\tilde{t}$ up to 6, within 2.5% for $\tilde{t}$ up to 6.20, but as big as 12% for $\tilde{t}=6.25$. To be able to continue the calculations with $N_{\Delta_1}=15360$, $N_{\Delta_2}=3840$ beyond $\tilde{t}=6$, we had to slightly modify the mesh stabilization method of Sec. III D in an empirical way, introducing the weight $[6 \times 10^{-5} \mathbf{x}_{ij}^{-2} + \mathbf{x}_{ij}^2]^2$ under the summation in the function (34). The continued elongation observed in the reliable range $\tilde{t} \leq 6$ (Fig. 19) strongly suggests eventual drop breakup. For a single drop sufficiently stretched in a linear flow field under supercritical conditions, it is well known^{29,30,50} that a sudden stop or reduction of the flow rate facilitates the drop fragmentation. Qualitatively, in our simulation, the smaller drop is submerged in some kind of extensional flow generated by the larger drop; however, the strength of this flow decreases very gradually as time proceeds and the drops depart, thus delaying the breakup. The global shape of the drop in Fig. 18(d) does show the very onset of the fragmentation, with the neck

forming, but further, more accurate calculations are necessary to confirm that this is not a numerical artifact. Our mesh stabilization method generates a grid close to optimal in the central part of the elongated drop (except for some nodes being too close), but is clearly non-adaptive in the higher mean curvature zone at the tip (Fig. 17); further work will reveal if an adaptive version of this method can be more successful in describing the details of the breakup. Most easily, the simulations in Fig. 18 could be continued assuming, with some approximation, the axial symmetry of the problem; however, the role of 3D disturbances (still present in Fig. 18), in the development of breakup is unclear.

Despite the difficulties in detailing the fragmentation, our code in the present form, with its very soft stability constraints, seems quite suitable for determining the critical values of the parameters Ca , a_1/a_2 , λ , and $\Delta\tilde{x}_0$ leading to breakup. Again, axisymmetrical codes^{29–33} can presumably be used to find the boundary in the parameter space (Ca , a_1/a_2 , λ) demarcating the values for which the breakup is possible. For each set of values (Ca , a_1/a_2 , λ) inside this boundary, 3D simulations similar to those in Fig. 17 can be used to determine the breakup efficiency, i.e., the maximum horizontal offset far upstream leading to breakup, by trial-and-error. This problem will be addressed in our future work. Finally, the breakup of a larger drop, a possible result of hydrodynamic interactions in gravity-induced motion for some values of the parameters, will be also investigated.

V. CONCLUDING REMARKS

We have developed a novel, three-dimensional boundary-integral algorithm for deformable drops moving in a viscous medium at low Reynolds numbers. The specific goal was to develop an efficient and accurate code for simulating the interaction of two drops in gravity-induced motion, and for further use in a systematic study of bubble and drop deformation-induced coalescence and breakup of sedimenting hydrodynamically interacting drops, which will be a direct continuation of the present work. However, our code is applicable to an imposed ambient flow and a larger number of drops; the direct generalization for the case of N drops in a periodic cell, with $O(N^2)$ scaling, may also be made (see Sec. I).

Our iterative method for the curvature and normal vector calculation on a triangulated surface, which is based on locally fitting the best paraboloid by least squares, was found to be considerably more robust and accurate in boundary-integral calculations than are contour integration schemes. We have also shown how the very poor convergence of traditional “successive substitutions” in the case of drops in very close approach with extremely small or large viscosity ratio λ can be dramatically improved by a novel strategy of combining biconjugate gradient and simple iterations. This strategy can be used both in deformation-induced bubble–bubble coalescence simulations ($\lambda \ll 1$), and in boundary-integral calculations of the interaction between solid ($\lambda = \infty$) non-spherical particles, to allow the surfaces to come sufficiently close that asymptotic lubrication forms⁵¹ can be used. Some more possible applications of this iterative method may include sedimentation and rheology simu-

lations for highly concentrated suspensions of rod-like (or disk-like) solid particles with arbitrary aspect ratio, when the slender-body approximation is not valid. Although we used a single-processor computer for all calculations, it is useful to note that the ideas of parallel computations developed for traditional iterations^{42,52} are equally applicable to the iterative strategy suggested in this work. Also a part of our code is a new and universal variational method for global mesh stabilization with very soft stability constraints. Finally, we have derived a curvatureless boundary-integral formulation and shown that, in the case of $\lambda = 1$ and well-separated (or single) drops, this formulation can be readily used for a more accurate simulation of the drop breakup than provided by the conventional boundary-integral formulation; further work is needed, however, to make this new formulation useful for $\lambda \neq 1$ and/or drops in close approach.

Several numerically challenging examples were selected to demonstrate that (i) our code is capable of bridging the gap between the boundary-integral calculations for finite deformations and the small capillary number case, where the asymptotic theory of close interaction and drop coalescence²¹ can be used, (ii) our code provides an accurate 3D simulation of the enhanced bubble coalescence phenomenon (for $\lambda \ll 1$) discovered by Manga and Stone,^{33,40} and (iii) hydrodynamic interactions of drops with $\lambda = O(1)$ in gravity-induced motion typically lead to drop breakup, even for moderate capillary numbers.

The advantages of the present mesh stabilization technique observed for buoyancy-driven drop interactions warrant additional discussion. Crude tests with $N_\Delta = 320$ elements per drop, and the scaling $\Delta\tilde{t} \sim N_\Delta^{-1/2}$, show that the initial time step $\Delta\tilde{t}_0$ for the grid tension method⁴¹ [even with the optimum coefficient ψ and different ways of constructing expression (33)] would need to be at least 27 times less than the stability limit for the current method in the breakup simulation (Figs. 17 and 18), to avoid eventual instability. Moreover, we found in a similar way that the bubble–bubble coalescence simulation (Fig. 13) would require the initial time step $\Delta\tilde{t}_0$ for the grid tension technique to be at least 60 times less than the stability limit for the present method. For slightly deformable drops with $\lambda = 1$ (Fig. 11), the maximum stability limit $\Delta\tilde{t}_0$ for the grid tension method was found to be closer to that for our mesh stabilization scheme (but still about five times less). In all of these examples, low resolution tests were also used to verify that vigorous tangential motions caused by grid tension not only make the stability constraints much tighter, but also distort the solutions for stable calculations compared to those for the present method at the same N_Δ ; the destructive effects of these tangential motions outweigh the advantage of mesh adaptation to near-contact and higher curvature zones. For these reasons, the approach based on grid tension, which was found successful for drops in shear flow,⁴¹ is not satisfactory for drops in gravity-induced motion, and the more complicated “passive” mesh stabilization method of Sec. III D is preferred. The passive method is currently being generalized to allow for mesh adaptation in near-contact and higher curvature zones; the latter is probably necessary for detailing the drop fragmentation and for very accurate calculation of critical

conditions leading to continued elongation and eventual breakup. It may prove very advantageous to use the new mesh stabilization method, even in its present form, for simulating sedimentation in highly concentrated emulsions of deformable drops, as long as the drop shape remains reasonably compact. We note finally that our, largely empirical time-step refinement strategy might not be optimal for very small separations (keeping the time steps well below the stability limit in this case), and so more sophisticated, adaptive time integrators are worth considering in future work.

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APPENDIX: THE DERIVATION OF (40) AND (41)

Let u^1, u^2 be the local surface parametrization on S_α , $\xi_i = \partial \mathbf{x} / \partial u^i$ the covariant basis vectors, $g_{ij} = \xi_i \cdot \xi_j$ the metric tensor, $\|g^{ij}\| = \|g_{ij}\|^{-1}$, and $\xi^i = g^{ij} \xi_j$ the contravariant basis vectors. The integral identity (21) projected on the ℓ th Cartesian axis with the unit vector $\mathbf{e}_\ell$ can be written as

$$\begin{aligned} 2 \int_S kn_\ell \sqrt{g} du^1 du^2 &= - \oint_C \mathbf{t} \cdot \mathbf{e}_\ell ds \\ &= - \int_S \nabla_S \cdot (\mathbf{e}_\ell) \sqrt{g} du^1 du^2, \end{aligned} \quad (\text{A1})$$

where C is a “small” closed contour (within the parametrization u^1, u^2), S is the surface portion inside C , $g = \det \|g_{ij}\|$ and $\sqrt{g} du^1 du^2$, as usual, is the surface area element. The surface divergence, $\nabla_S \cdot (\mathbf{e}_\ell)$, can be calculated by the general rule,

$$\nabla_S \cdot (\mathbf{e}_\ell) = g^{-1/2} \frac{\partial (\mathbf{e}_\ell \cdot \xi^i g^{1/2})}{\partial u^i}. \quad (\text{A2})$$

Using (A2) yields the differential form of (A1),

$$2k\mathbf{n}\sqrt{g} = - \frac{\partial}{\partial u^i} (\sqrt{g} \xi^i), \quad (\text{A3})$$

which is equivalent to the usual²³ $\nabla \cdot \mathbf{n} = 2k$, but is more convenient in the present context. The identity (A3) results in

$$2k\mathbf{n} \cdot \mathbf{G}^{(\ell)}(\mathbf{x} - \mathbf{y}) = \xi^i \cdot \frac{\partial \mathbf{G}^{(\ell)}}{\partial u^i} - \nabla_S \cdot (\mathbf{G}^{(\ell)}) \quad (\text{A4})$$

for Green’s function (4) [obviously, the first term on the right-hand side of (A4) is an invariant independent of the local parametrization]. When $\mathbf{y}$ is not on S_α , the relation (A4) can be integrated over the whole surface S_α , yielding

$$\int_{S_\alpha} 2k\mathbf{n} \cdot \mathbf{G}^{(\ell)}(\mathbf{x} - \mathbf{y}) dS_x = \int_{S_\alpha} \xi^i \cdot \frac{\partial \mathbf{G}^{(\ell)}}{\partial u^i} dS_x. \quad (\text{A5})$$

When $\mathbf{y} \in S_\alpha$, a surface portion lying inside a sphere of a small radius ϵ and centered at $\mathbf{y}$ is excluded from the inte-

gration; in this case, however, the flux of $\mathbf{G}^{(\ell)}$ through the small surface contour around $\mathbf{y}$ vanishes, when $\epsilon \rightarrow 0$ [as the explicit form (4) shows], and so (A5) still holds, if the right-hand side is a principal-value integral.

By the chain rule,

$$\xi^i \cdot \frac{\partial \mathbf{G}^{(\ell)}}{\partial u^i} = \frac{\partial G_{\ell m}}{\partial x^n} (\xi^i)_m (\xi_i)_n, \quad (\text{A6})$$

where the subscripts m and n denote the Cartesian coordinates of vectors. The relation $\xi^i = g^{ij} \xi_j$ and the symmetry of g^{ij} show that $\partial G_{\ell m} / \partial x^n$ in (A6) can be replaced by the rate-of-strain tensor

$$E_{mn}^{(\ell)} = \frac{1}{2} \left(\frac{\partial G_{\ell m}}{\partial x^n} + \frac{\partial G_{\ell n}}{\partial x^m} \right), \quad (\text{A7})$$

and (A6) can be further transformed to

$$\xi^i \cdot \frac{\partial \mathbf{G}^{(\ell)}}{\partial u^i} = \frac{1}{2} (q^{(\ell)} \delta_{mn} + \tau_{mn}^{(\ell)}) (\xi^i)_m (\xi_i)_n, \quad (\text{A8})$$

where $q^{(\ell)}$ and $\tau_{mn}^{(\ell)}$ are the pressure (8) and the stress tensor (6) for the fundamental solution. The identity $\xi^i \cdot \xi_i = 2$ and the explicit expressions (8) and (6) yield

$$\xi^i \cdot \frac{\partial \mathbf{G}^{(\ell)}}{\partial u^i} = \frac{1}{8\pi} \left[\frac{3(\mathbf{r} \cdot \xi^i)(\mathbf{r} \cdot \xi_i) r_\ell}{r^5} - \frac{2r_\ell}{r^3} \right]. \quad (\text{A9})$$

In (A9), $(\mathbf{r} \cdot \xi^i)(\mathbf{r} \cdot \xi_i)$ can be recognized as the square of the length of the tangential vector $\mathbf{r}^\parallel = \mathbf{r} - [\mathbf{r} \cdot \mathbf{n}(\mathbf{x})] \mathbf{n}(\mathbf{x})$, and so (A5) takes the final form (40).

To prove (41), we note that

$$\begin{aligned} \frac{\mathbf{r} \cdot \mathbf{n}(\mathbf{y})}{r^3} \left\{ 1 - \frac{3[\mathbf{r} \cdot \mathbf{n}(\mathbf{x})]^2}{r^2} \right\} \\ = \frac{\mathbf{r} \cdot [\mathbf{n}(\mathbf{x}) + \mathbf{n}(\mathbf{y})]}{r^3} \left\{ 1 - \frac{3[\mathbf{r} \cdot \mathbf{n}(\mathbf{x})][\mathbf{r} \cdot \mathbf{n}(\mathbf{y})]}{r^2} \right\} \\ + 4\pi n_\ell(\mathbf{y}) n_n(\mathbf{y}) \tau_{nm}^{(\ell)}(\mathbf{x} - \mathbf{y}) n_m(\mathbf{x}) + \frac{\partial(1/r)}{\partial \mathbf{n}(\mathbf{x})}. \end{aligned} \quad (\text{A10})$$

When $\mathbf{y}$ is not on S_α , the last two terms in (A10) do not contribute to the integral over S_α ; when $\mathbf{y} \in S_\alpha$, their contributions, $2\pi \delta_{\ell n} n_\ell(\mathbf{y}) n_n(\mathbf{y})$ and -2π , cancel, and so (41) also holds.

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